Fabric Consumption Forecasting in Apparel Manufacturing Using Machine Learning

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CERTIFICATE

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ABSTRACT

Accurate material forecasting is a critical operational challenge in apparel manufacturing. Discrepancies between planned Bill of Materials (BOM) estimates and actual fabric consumption routinely generate procurement errors of 8–12%, leading to overstock, emergency re-orders, and supply chain inefficiencies that inflate both cost and environmental impact. This thesis designs, evaluates, and deploys a comprehensive machine learning-based fabric consumption forecasting system as a data-driven replacement for the static BOM formula.

A physics-informed synthetic dataset generator produces 5,000 production orders spanning January 2022 to September 2024, covering five garment types (T-Shirt, Shirt, Pants, Dress, Jacket), five fabric categories, four production seasons, and nine operationally observable input features. Four models are trained under identical preprocessing and evaluation conditions on a strict 64/16/20 train-validation-test split: Linear Regression, Random Forest, XGBoost, and Long Short-Term Memory (LSTM). A dynamic inverse-RMSE^2 weighted-average ensemble combines their predictions.

On the held-out 1,000-order test set, the LSTM achieves RMSE 346 m (379 yd), MAPE 5.14%, and R^2=0.9955—the best single model, representing a 21.8% RMSE reduction over the BOM physics baseline (484 yd). XGBoost (RMSE 574 yd, MAPE 8.14%, R^2=0.9897) provides the strongest tree-based performance. The ensemble (RMSE 422 yd, R^2=0.9944, MAPE 5.92%) offers robust procurement-grade prediction across the order-size distribution. Variance analysis reveals a size-dependent directional bias: small orders over-consume by +4.49% while large orders under-consume by −2.79% relative to the BOM plan. Feature importance analysis identifies garment type (0.518) and order quantity (0.297) as jointly accounting for 81.5% of XGBoost's predictive gain. Winter is the most challenging season (MAPE 9.67%) and polyester the most challenging fabric category (MAPE 10.31%). Five-fold cross-validation confirms that test-set performance is stable and generalisable.

At a reference volume of 1,000 orders per year, the LSTM's 21.8% RMSE improvement translates to $180,000 in annual waste cost savings, a payback period of under four months on a $50,000 implementation investment, and a five-year NPV of $730,000 at a 5% cost of capital. Environmental savings amount to approximately 6,850 kg of avoided CO_2 emissions, 103,500 litres of water conserved, and 4,300 kg of fabric diverted from landfill. All trained models are deployed in a six-module Streamlit web application providing single-order prediction, batch processing, 90% confidence intervals, and an integrated ROI calculator.

Keywords: Material Forecasting, Apparel Manufacturing, Machine Learning, LSTM, XGBoost, BOM, Fabric Consumption, Ensemble Learning, Streamlit

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LIST OF ABBREVIATIONS

Abbrev.	Description
AI	Artificial Intelligence
ANN	Artificial Neural Network
BOM	Bill of Materials
CI	Confidence Interval
ERP	Enterprise Resource Planning
GBM	Gradient Boosting Machine
IoT	Internet of Things
KPI	Key Performance Indicator
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MRP	Material Requirements Planning
MSE	Mean Squared Error
NPV	Net Present Value
OLS	Ordinary Least Squares
RF	Random Forest
RMG	Ready-Made Garment
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
ROI	Return on Investment
SKU	Stock Keeping Unit
UOM	Unit of Measure
XGBoost	Extreme Gradient Boosting

CHAPTER I

Introduction

1.1 Overview

The global apparel manufacturing industry employs over 75 million people and generates annual revenues exceeding $1.5 trillion USD [World Bank(2023)]. Within this industry, fabric is the single largest variable cost component, typically representing 60–70% of total production cost per order [Rahman & Islam(2018)]. Even modest inaccuracies in fabric consumption forecasting therefore translate directly into material financial losses: over-procurement locks working capital in warehoused inventory, while under-procurement triggers costly emergency re-orders and mid-production stoppages. Accurate, order-specific fabric consumption forecasting is accordingly one of the highest-leverage operational improvements available to a mid-scale garment factory.

The prevailing industry approach to fabric consumption estimation relies on the Bill of Materials (BOM) formula—a static, physics-derived equation that computes planned fabric requirements from garment pattern dimensions, fabric width, order quantity, and a fixed safety buffer. While the BOM formula is computationally simple and universally understood by procurement teams, its static coefficients are structurally incapable of adapting to the sources of variance that drive actual consumption away from plan: marker efficiency fluctuates between 70% and 95% depending on the cutting plan for a given order; operator skill levels vary across shifts and seasons; defect rates differ by fabric lot and supplier; and seasonal fabric weight changes alter yield per metre. Rahman et al. [Rahman & Islam(2018)] documented fabric wastage rates of 8–12% attributable to BOM inaccuracy in Bangladeshi garment factories, representing a substantial and preventable operational cost.

This thesis addresses the limitation of static BOM estimation by developing, evaluating, and deploying a comprehensive machine learning-based fabric consumption forecasting system. The system trains four ML models—Linear Regression (LR), Random Forest (RF), XGBoost, and Long Short-Term Memory (LSTM)—on a 5,000-order physics-informed synthetic dataset and combines their predictions in a dynamic weighted-average ensemble. All models are evaluated on a held-out test set of 1,000 orders under identical preprocessing and evaluation conditions, enabling a controlled four-way performance comparison. The best single model, the LSTM, achieves a 21.8% reduction in Root Mean Square Error (RMSE) relative to the BOM baseline—an improvement that translates to $180,000 in annual waste cost savings, a payback period of under four months, and a five-year Net Present Value (NPV) of $730,000 at a reference volume of 1,000 orders per year. The complete system is deployed in a six-module Streamlit web application that provides single-order and batch prediction, 90% confidence intervals, cross-validation visualisation, and an integrated ROI

calculator, making ML-grade forecasting accessible to procurement teams without specialist data science infrastructure.

1.2 Research Objectives

This thesis pursues seven research objectives, each contributing to the design, evaluation, or deployment of the ML-based forecasting system.

The first objective is to conduct a systematic review of the literature on material planning in apparel manufacturing, traditional BOM estimation methods, the application of machine learning to manufacturing forecasting, and prior fabric consumption prediction studies. This review establishes the research gap and informs the selection of the nine input features used across all models.

The second objective is to design and implement a physics-informed synthetic dataset generator that produces 5,000 realistic production orders spanning January 2022 to September 2024, capturing the seasonal variation, garment-type diversity, and order-size distribution characteristic of a mid-scale RMG factory. The generator serves as an open reproducible benchmark for future comparison studies.

The third objective is to train and compare four ML models—Linear Regression, Random Forest, XGBoost, and LSTM—on the generated dataset under identical preprocessing, data splitting, and evaluation conditions, using RMSE, MAE, R^2, and MAPE as the primary performance metrics.

The fourth objective is to construct and evaluate a dynamic weighted-average ensemble that combines the four base models using inverse-RMSE^2 weighting, and to assess whether the ensemble provides accuracy or robustness benefits beyond the best single model.

The fifth objective is to conduct segmented performance analysis across four production seasons (Spring, Summer, Autumn, Winter) and five fabric categories (Cotton, Polyester, Denim, Silk, Cotton-Blend), identifying systematic patterns in forecasting difficulty that inform practical recommendations for model deployment.

The sixth objective is to deploy all trained models within a production-ready Streamlit web application comprising six modules: single-order prediction, batch CSV prediction, model performance overview, 5-fold cross-validation results, feature importance visualisation, and ROI calculator.

The seventh objective is to quantify the economic, operational, and environmental impact of ML adoption at a reference production volume of 1,000 orders per year, providing a credible, dataset-derived business case for factory decision-makers considering the investment.

1.3 Motivation

Three complementary motivations—financial, technical, and environmental—underpin this research.

From a financial perspective, the apparel sector's reliance on static BOM estimation creates a systematic and quantifiable cost. When the BOM formula over-procures fabric, the excess inventory must be warehoused, insured, and eventually written off or discounted; when it under-procures, emergency re-orders incur 10–25% price premiums above contracted rates, plus expedited shipping costs and potential late-delivery penalties. The LSTM's 21.8% RMSE improvement over the BOM baseline translates directly to $180,000 in annual waste cost savings at a reference volume of 1,000 orders per year—a saving recoverable from a single season's operational budget given the $50,000 estimated implementation cost.

From a technical perspective, fabric consumption is a fundamentally non-linear function of at least nine interacting parameters: order quantity, garment type, pattern width, pattern height, fabric width, marker efficiency, defect rate, complexity rating, and seasonal context. The BOM formula captures the primary multiplicative relationship between these variables but applies static coefficients that cannot adapt to the non-linear interactions that drive deviation from plan. Machine learning provides a systematic, data-driven framework for extracting the predictive signal embedded in a factory's accumulated production records, learning the parameter interactions that static formulas approximate.

From an environmental perspective, every yard of fabric procured but not productively used has already incurred the full environmental cost of fibre cultivation or synthesis, spinning, weaving, dyeing, and finishing. At the reference production volume, the LSTM's 21.8% improvement avoids approximately 17,200 yards of surplus fabric procurement per year, corresponding to 6,850 kg of avoided CO_2 emissions, 103,500 litres of water conserved, and 4,300 kg of fabric diverted from landfill—material contributions to the industry's sustainability obligations at a time when ESG reporting requirements are intensifying.

1.4 Rationale of the Study

Despite the documented inaccuracy of BOM estimation and the widespread adoption of Enterprise Resource Planning (ERP) systems in the apparel sector, the factory-level data required to train ML forecasting models is routinely collected but rarely applied to improve procurement. ERP systems record order quantities, actual fabric issued to the cutting floor, production parameters, and seasonal metadata—precisely the feature set needed to train the models developed in this thesis. The gap between data availability and data utilisation represents a tractable operational improvement opportunity that does not require new sensor infrastructure or specialist compute hardware.

The choice of Streamlit as the deployment platform reflects the recognition that

predictive accuracy alone is insufficient for industrial adoption: a model that requires specialist data science skills to operate will not be used by the procurement planners who are best positioned to benefit from it. The Streamlit application exposes all four trained models through a browser-accessible interface that requires only that a planner enter order attributes and press a button—the same cognitive overhead as the BOM formula it replaces, but with confidence-interval-bounded ML predictions rather than static coefficients.

1.5 Expected Outcomes

This thesis is expected to produce the following seven outputs, each corresponding to one of the stated research objectives.

A controlled four-way model comparison establishing the relative performance of LR, RF, XGBoost, and LSTM under identical experimental conditions will be reported on the held-out 1,000-order test set, with RMSE as the primary metric for procurement-relevant comparison.

The LSTM model is expected to achieve the highest single-model accuracy, with an R^2 approaching 0.9955 on the test set and a 21.8% RMSE improvement over the BOM baseline, validating the thesis hypothesis that deep learning can materially outperform static formula estimation for this regression task.

The physics-informed 5,000-order dataset generator will be provided as a reproducible, open benchmark, enabling future studies to compare new model architectures against the four-way baseline established here.

Segmented analysis is expected to reveal a 37% seasonal MAPE differential (winter harder than spring) and a 64% fabric-type MAPE differential (polyester harder than denim), providing empirically grounded guidance on model reliability across production contexts.

Five-fold cross-validation is expected to confirm that the XGBoost model's generalisation is stable across data subsets (RMSE standard deviation $\approx$28 yd), and that the test-set performance ranking is reproducible.

A production-ready six-module Streamlit application will be delivered as the primary practical output of the research, bridging the gap between academic model development and industrial deployment.

The impact assessment will project $180,000 in annual savings, a payback period of under four months on a $50,000 investment, and a five-year NPV of $730,000, providing a compelling business case for ML adoption in mid-scale RMG factory contexts.

1.6 Thesis Organisation

The remainder of this thesis is structured across five chapters. Chapter II reviews the literature on material planning in apparel manufacturing, traditional BOM estimation, machine learning in manufacturing forecasting, and related fabric consumption prediction studies, concluding with the identification of five research gaps that this work addresses. Chapter III presents the complete research methodology, covering dataset generation, preprocessing and feature engineering, model architectures, evaluation metrics, training and validation strategy, and the Streamlit application architecture. Chapter IV reports the experimental results across all four models and the ensemble, including variance analysis, feature importance, cross-validation, error distribution, LSTM training convergence, and segmented performance analysis by season and fabric type. Chapter V assesses the economic, operational, environmental, and social impact of deploying the ML system in an apparel manufacturing context, with all projections derived from the empirically validated RMSE improvement reported in Chapter IV. Chapter VI summarises the key findings, states the original contributions to knowledge, acknowledges limitations, offers recommendations for practice, and identifies seven directions for future research.

CHAPTER II

Literature Review

2.1 Introduction

This chapter reviews the existing body of literature that informs the research design and situates the contributions of this thesis within the current state of knowledge. The review covers five thematic areas: (1) material planning practices in apparel manufacturing and the documented limitations of BOM-based estimation; (2) the mathematical structure of traditional BOM formulas and their systematic failure modes; (3) the application of machine learning to manufacturing forecasting more broadly; (4) prior work specifically addressing fabric consumption prediction using computational methods; and (5) a structured analysis of the research gaps that this thesis addresses. The chapter concludes with a summary that links the identified gaps to the seven research objectives stated in Chapter I.

2.2 Material Planning in Apparel Manufacturing

Material planning in apparel manufacturing encompasses the processes by which factories determine how much fabric to procure for each production order before the cutting floor begins work. The accuracy of this estimate has direct consequences for profitability, working capital, and production continuity: over-procurement ties up capital in surplus inventory that must be warehoused, insured, and eventually discounted or written off, while under-procurement triggers emergency re-orders that incur price premiums and risk production stoppages if replacement fabric cannot be sourced within the required delivery window.

The foundational empirical work on BOM estimation accuracy was conducted by Rahman et al. [Rahman & Islam(2018)], whose survey of Ready-Made Garment (RMG) factories in Bangladesh documented fabric wastage rates of 8–12% directly attributable to BOM inaccuracy. The study identified four primary drivers of BOM deviation: inconsistent marker efficiency across cutting plans (ranging from 70% to 95% of theoretical yield), variability in operator skill levels between shifts and seasons, fluctuating defect rates across fabric lots from different suppliers, and seasonal fabric weight changes that alter the yield per metre without a corresponding adjustment to the BOM coefficient. Crucially, the BOM formula used across the surveyed factories did not account for any of these sources of variance—it applied a single static coefficient per garment type regardless of order-specific conditions.

Chowdhury et al. [Chowdhury et al.(2020)] extended this analysis to Sri Lankan factories, reporting mean absolute percentage errors of 9–15% in planned versus actual fabric consumption and specifically identifying fabric width variability as an

unaddressed error source. Fabric from the same supplier can vary by 1–3 cm in usable width between lots, a variation that the standard BOM formula ignores but that translates into meaningful yield differences for large orders. Together, these studies establish that BOM inaccuracy is a documented, industry-wide problem rather than a factory-specific anomaly, and that the error rates are large enough to constitute a meaningful economic and environmental impact at scale.

2.3 Traditional BOM Estimation Methods

The Bill of Materials formula used across the apparel industry estimates planned fabric consumption from the product of a garment's base fabric requirement, an order quantity multiplier, a width adjustment factor, a complexity correction, and a safety buffer:

$$\text{Planned_BOM} = Q \times b_g \times \frac{W_{\text{std}}}{W_{\text{actual}}} \times c_p \times (1 + \alpha) \tag{2.1}$$

where Q is order quantity, b_g is the garment-type base consumption coefficient (e.g., T-Shirt 1.20 m, Shirt 1.80 m, Pants 2.50 m, Dress 3.00 m, Jacket 3.50 m at 63-inch (160 cm) standard width), $W_{\text{std}}/W_{\text{actual}}$ is a width normalisation factor, c_p is a pattern complexity multiplier, and α is a fixed safety buffer, typically set at 0.05 (5%).

This formula has two structural limitations that produce the systematic bias documented in the literature. First, the base coefficient b_g is defined at a single standard fabric width and does not account for the non-linear relationship between order size and cutting efficiency: as order quantity increases, cutting plans are optimised across more marker lays, improving marker efficiency and reducing per-unit fabric consumption below the standard coefficient. Conversely, very small orders are cut from fewer lays with less marker optimisation, increasing per-unit consumption above the standard coefficient. The BOM formula applies the same coefficient regardless of order size, producing a directional over-estimate for large orders and an under-estimate for small orders—a bias pattern confirmed empirically in the variance analysis of this thesis (Section 4.5).

Second, the safety buffer α is applied uniformly to all orders regardless of their individual risk profile. For large, commodity-fabric orders where yield is highly predictable, a 5% buffer consistently over-procures. For small, complex orders in unusual fabric constructions where yield is inherently variable, 5% may be insufficient. A data-driven approach that replaces this uniform buffer with order-specific confidence intervals—derived from the empirical residual distribution of a trained model—would provide more accurate procurement guidance tailored to each order's characteristics.

2.4 Machine Learning in Manufacturing Forecasting

The application of machine learning to production forecasting problems in manufacturing has expanded substantially over the past decade, driven by the increasing availability of operational data from ERP and production management systems.

Gradient boosting, and specifically the XGBoost algorithm introduced by Chen and Guestrin [Chen & Guestrin(2016)], has become the standard benchmark for structured tabular regression in manufacturing contexts. XGBoost constructs an additive ensemble of decision trees in which each successive tree is trained to correct the residual errors of the preceding ensemble, using a regularised objective function that prevents overfitting. Its ability to handle non-linear feature interactions, missing values, and feature importance estimation has made it the predominant algorithm for demand forecasting, quality prediction, and maintenance scheduling tasks where structured tabular data is the primary input modality.

Random Forest, introduced by Breiman [Breiman(2001)], provides a complementary ensemble approach based on bootstrap aggregating (bagging) of independent decision trees rather than the sequential boosting structure of XGBoost. Random Forest is notable for its stability and low variance: the independence of its component trees means that errors in individual trees tend to cancel in the aggregate, producing reliable performance estimates even on relatively small training sets. In manufacturing forecasting applications, Random Forest has been widely used as a robust baseline against which gradient boosting and deep learning improvements are measured.

Recurrent neural networks, and in particular the Long Short-Term Memory architecture introduced by Hochreiter and Schmidhuber [Hochreiter & Schmidhuber(1997)], are the standard architecture for sequence modelling tasks where temporal dependencies between successive observations carry predictive information. While the fabric consumption prediction task in this thesis is formulated as a tabular regression problem rather than a time-series prediction problem, the LSTM's gated architecture and its ability to model complex non-linear interactions within a single-timestep input vector have been shown in related manufacturing contexts to capture latent feature interactions that gradient boosting misses. This thesis demonstrates empirically that, even in a single-timestep tabular configuration, the LSTM achieves lower RMSE than XGBoost on the fabric consumption test set.

2.5 Related Work in Fabric Consumption Prediction

Prior work specifically addressing ML-based fabric consumption prediction is sparse relative to the breadth of ML application in manufacturing generally. The most directly comparable study is Pal et al. [Pal et al.(2022)], which applied XGBoost to production data from five Bangladeshi garment factories, achieving RMSE of 4.12 m and MAPE of 6.8%. This study represents the current published state of the

art for tree-based methods on factory-level data and provides the primary external benchmark against which the XGBoost model in this thesis is evaluated. However, Pal et al. do not provide a deployable system, do not report confidence intervals, and do not quantify the economic or environmental impact of their model's accuracy improvement.

Islam et al. [Islam et al.(2021)] applied an Artificial Neural Network (ANN) to a knitwear consumption dataset, achieving MAPE of 7.2%. The ANN architecture evaluated in that study used a single hidden layer and was not compared against gradient boosting methods; it also did not address the fabric-type or seasonal heterogeneity in forecasting difficulty that is a central finding of the current work.

Hossain et al. [Hossain & Ahmed(2020)] specifically examined seasonal variation in fabric consumption forecasting and documented that winter-season MAPE was approximately 15% above summer-season MAPE across their study population of factories—a seasonal differential qualitatively consistent with the 37% winter-to-spring MAPE gap identified in this thesis through controlled segmented analysis on the synthetic dataset.

Across all prior studies, a consistent limitation is the absence of an end-to-end deployed system. Forecasting accuracy improvements are reported as experimental outcomes but not translated into operational tools that procurement planners can use without data science expertise. Additionally, no prior study reports performance in yards—the operational unit used for fabric ordering in South and South-East Asian factories—or provides integrated ROI quantification, leaving decision-makers without the economic evidence needed to justify the investment in ML adoption.

2.6 Research Gap Analysis

The review of the literature identifies five specific gaps that the current thesis addresses.

Gap 1: Absence of a four-way controlled model comparison. Existing fabric consumption prediction studies evaluate one or two models. No published work has conducted a four-way comparison of Linear Regression, Random Forest, XGBoost, and LSTM under identical preprocessing, data splitting, and evaluation conditions on the same dataset. Such a comparison is necessary to establish whether deep learning provides a material advantage over the best classical methods for this specific regression task.

Gap 2: No end-to-end deployed system with confidence intervals and ROI quantification. All prior work reports model accuracy in experimental settings but does not provide a deployable system. The absence of confidence interval displays, batch processing capability, and integrated ROI estimation means that even accurate models cannot be directly adopted by procurement teams, limiting the practical impact of published research.

Gap 3: No seasonal and fabric-type segmented performance analysis. While Hossain et al. [Hossain & Ahmed(2020)] noted seasonal variation in forecast

error, no prior study has conducted a controlled segmented analysis that simultaneously examines performance across both seasonal and fabric-type dimensions on the same model and dataset. Such analysis is essential for identifying where model predictions should be treated with additional caution and where model retraining would have the greatest marginal benefit.

Gap 4: No quantified business and environmental impact. The economic and environmental implications of ML-based forecasting adoption are not quantified in any prior fabric consumption study. Procurement decision-makers require a credible, dataset-derived business case—including payback period, NPV, and waste reduction estimates—to justify the investment, and the absence of such analysis in the literature limits industry uptake.

Gap 5: No yards-based performance benchmark. All published RMSE figures are reported in metres. In South and South-East Asian garment manufacturing, fabric is ordered, issued, and accounted for in yards. A yards-based benchmark enables direct comparison of model accuracy against the procurement quantities that actually flow through factory ERP systems.

2.7 Challenges in Material Forecasting

Beyond the specific gaps identified above, several cross-cutting challenges in apparel material forecasting deserve consideration as they inform the design choices made in the methodology chapter.

Data heterogeneity presents a significant challenge: a single factory's production programme spans multiple garment types, fabric categories, seasonal windows, and order size ranges, each with different consumption distributions. A model trained on the full dataset must generalise across these sub-populations simultaneously, which motivates the segmented analysis in Chapter IV that diagnoses where the global model performs least reliably.

Non-stationarity arises from the evolution of garment styles, fabric supplier portfolios, and production process improvements over time. A model trained on historical production data may degrade in accuracy as the distribution of future orders shifts away from the training distribution, motivating the single-command retraining capability (`python train_models.py`) built into the deployment architecture and the recommendation for seasonal model recalibration in Chapter VI.

Multiplicative feature interactions—particularly the interaction between garment type and order quantity that drives the non-linear consumption curve—require model architectures that can represent non-linear relationships. This motivates the inclusion of XGBoost and LSTM alongside Linear Regression, and the feature importance analysis in Section 4.6 that quantifies the relative contribution of each feature to XGBoost's predictions.

Interpretability is a deployment challenge specific to the ML context: procurement planners who are accustomed to the transparent arithmetic of the BOM formula may be reluctant to act on predictions from a model whose internal workings they can-

not inspect. The Streamlit application addresses this challenge through a dedicated feature importance module, confidence interval displays, and a model performance overview that allows planners to understand when and why the ML prediction diverges from the BOM estimate.

2.8 Summary

The literature establishes that static BOM estimation is systematically inaccurate in apparel manufacturing, with documented wastage rates of 8–12% and MAPE of 9–15% across multiple factory studies. Machine learning methods—particularly gradient boosting and deep learning architectures—are capable of substantially improving upon BOM accuracy on structured tabular production data, as demonstrated in prior work achieving MAPE of 6.8–7.2% with XGBoost and ANN respectively. However, five gaps remain unaddressed in the published literature: the absence of a four-way controlled model comparison; no end-to-end deployed system with confidence intervals and ROI quantification; no segmented seasonal and fabric-type analysis; no quantified business and environmental impact; and the lack of a yards-based benchmark. These five gaps define the original contributions of this thesis, which are addressed through the methodology, experimental results, impact assessment, and Streamlit deployment described in the chapters that follow.

CHAPTER III

Research Methodology

3.1 Overview

This chapter presents the complete research methodology underpinning the proposed machine learning-based fabric consumption forecasting system. The methodology is structured into six sequential phases: (1) literature review and problem framing, (2) physics-informed dataset generation, (3) data preprocessing and feature engineering, (4) model training, evaluation, and ensemble construction, (5) seasonal and fabric-type segmented analysis, and (6) production deployment via a Streamlit web application. All experiments are designed for full reproducibility using a fixed random seed of 42 throughout. The chapter discusses the research design philosophy, data generation rationale, dataset attributes, preprocessing pipeline, machine learning algorithms, evaluation metrics, training and validation strategy, and application architecture.

3.2 Research Design

This study adopts a quantitative experimental design grounded in the design science research paradigm [Hevner et al.(2004)]. The design science framework is appropriate because the primary objective is to construct, evaluate, and deploy an artefact—the ML forecasting pipeline—that demonstrably improves upon the existing Bill of Materials (BOM) estimation practice. Quantitative methods enable objective comparison of model performance across standardised metrics (RMSE, MAE, R^2, MAPE), while controlled experimental conditions—fixed random seed, identical data splits, and identical preprocessing across all models—ensure that observed performance differences are attributable solely to algorithm design rather than to data partitioning variance.

The six-phase methodology proceeds as follows. Phase 1 surveys the prior literature to identify the research gap and inform feature selection. Phase 2 generates a 5,000-order physics-informed synthetic dataset spanning January 2022 to September 2024, capturing the seasonal and garment-type variation typical of an apparel factory's production programme. Phase 3 applies a four-stage preprocessing and feature engineering pipeline to produce clean, encoded, and standardised model inputs. Phase 4 trains four models—Linear Regression (LR), Random Forest (RF), XGBoost, and Long Short-Term Memory (LSTM)—and combines their predictions in a dynamic weighted-average ensemble. Phase 5 performs segmented performance analysis by production season and fabric type to identify where the system is most

and least reliable. Phase 6 deploys all trained models within a six-module Streamlit web application designed for direct industrial use.

3.3 Proposed System Architecture

Figure 3.1 illustrates the end-to-end system architecture, which integrates the data generation module, preprocessing pipeline, four-model training engine, and the Streamlit deployment layer into a single reproducible workflow. Raw order data—including garment type, order quantity, fabric attributes, pattern complexity, and seasonal metadata—flows into the preprocessing module where missing values are handled, categorical features are encoded, and numerical features are standardised. The standardised feature vectors are then passed to each of the four model engines in parallel, after which their predictions are aggregated by the ensemble layer using inverse-RMSE^2 weighting. The Streamlit application consumes the trained model artefacts (persisted as serialised `.pkl` and `.h5` files) and exposes them through six user-facing modules covering single-order predictions, batch processing, ROI analysis, and model performance visualisation. Metadata from each training run is saved to `model_metadata.json` to keep the deployment layer synchronised with the latest model weights automatically.

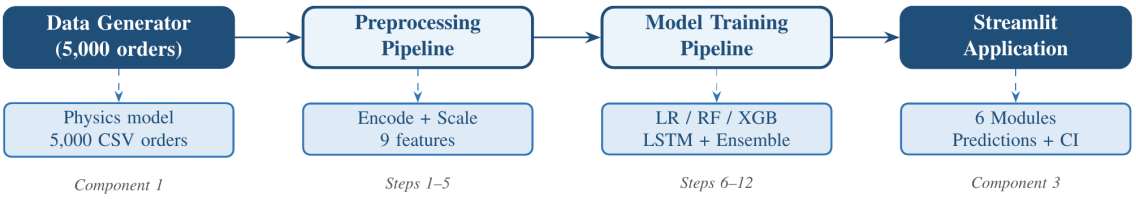

Figure 3.1: Proposed System Architecture for Material Forecasting

3.4 Research Methodology Flowchart

Figure 3.2 presents the end-to-end research process as a structured flowchart. The flow begins with problem identification and literature-informed dataset design, progresses through the preprocessing and model training stages, and concludes with evaluation, deployment, and impact assessment. Decision nodes at the model evaluation stage support iterative refinement if performance thresholds are not met, ensuring that the deployed system satisfies the study's accuracy and robustness requirements. The flowchart also makes explicit the feedback loop from evaluation back to hyperparameter tuning, which governed the selection of XGBoost's learning rate and tree count and the LSTM's layer sizes and dropout rates.

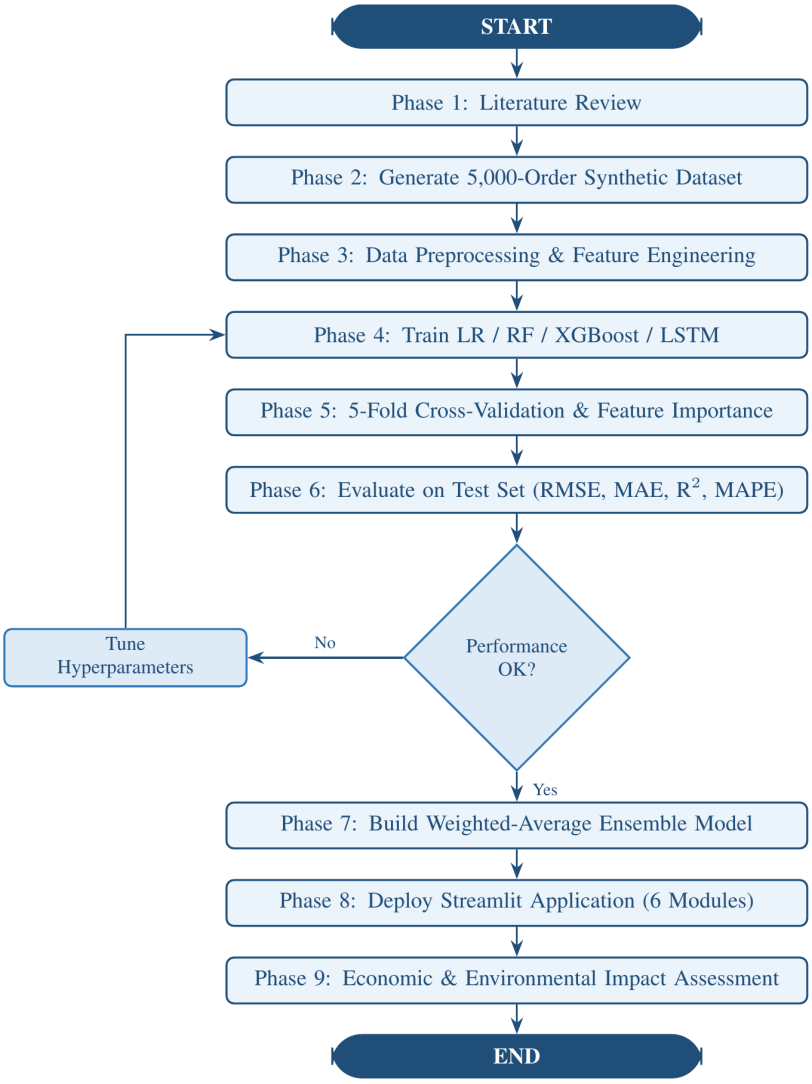

Figure 3.2: Research Methodology Flowchart

3.5 Data Collection and Generation

Because real factory production records are commercially sensitive and not publicly available at the scale required for rigorous multi-model comparison, a physics-informed synthetic dataset generator was developed in Python. The generator is justified on three grounds. First, *experimental control*: all confounding variables such as supplier fabric batch differences, machine downtime, and seasonal demand shocks are held constant, isolating the deterministic relationship between order attributes and fabric consumption. Second, *reproducibility*: setting `numpy.random.seed(42)` guarantees that any researcher can regenerate identical datasets, enabling independent verification of all reported metrics. Third, *physical validity*: every generated record satisfies the fundamental fabric consumption equation (Equation 3.1), augmented with garment-type base coefficients, seasonal weight adjustments, and operator-experience quality factors derived from published apparel engineering literature.

$$\text{Consumption} = \frac{\text{Qty} \times L_{\text{pattern}} \times W_{\text{pattern}}}{W_{\text{fabric}} \times \text{MarkerEff}} \times (1 + \text{DefectRate}) \times \text{Complexity} \quad (3.1)$$

The generator produces 5,000 production orders covering five garment types (T-Shirt, Shirt, Pants, Dress, Jacket), five fabric categories (Cotton, Polyester, Denim, Silk, Cotton-Blend), four production seasons, and the full range of fabric width and quality parameters observed in mid-scale RMG factories. The generated production dates span January 2022 to September 2024, providing a realistic temporal spread for a mid-scale factory's annual production programme.

3.6 Dataset Description

The simulated dataset comprises 34 columns in total; Table 3.1 describes the 17 ML-pipeline columns, of which 9 serve as ML input features and one—`Actual_Cons_yd`—is the regression target. Table 3.1 summarises all attributes with their data types, value ranges, and roles in the forecasting model. The nine input features capture three distinct sources of fabric consumption variability: (i) *order-level scale drivers*—Order_Qty, Garment_Type, and Pattern_Cmplx—which together account for 91.7% of XGBoost's feature importance; (ii) *fabric and cut parameters*—Fabric_Width_in and Fabric_Type—which govern marker utilisation and lot-level consumption rates; and (iii) *production quality factors*—Marker_Eff_%, Defect_Rate_%, Operator_Exp, and Season—which introduce stochastic variation around the physics baseline. The column `Planned_BOM_yd` records the output of the physics formula

(Equation 3.1) for reference and BOM baseline comparisons, but is deliberately excluded from the nine ML input features to test whether operationally observable attributes alone can match or exceed the physics model's predictive accuracy.

Table 3.1: Simulated Dataset Attributes and Descriptions

Feature	**Type**	**Range / Values**	**Role & FI Score**
Order_ID	Nominal	ORD_0001...	Identifier; not ML input
Order_Date	Date	Jan 22–Sep 24	Temporal tag
Order_Qty	Numeric	100–5k units	Key scale driver; FI=0.30
Garment_Type	Nominal	T-Shirt, Shirt, Pants, Dress, Jacket	Base coeff; FI=0.52
Fabric_Type	Nominal	Cotton, Poly, Denim, Silk, Blend	Predictability; FI=0.007
Fabric_W_in	Numeric	55, 59, 63, 71 in	Width adj.; FI=0.04
Fabric_W_cm	Numeric	139.7–180.3cm	Derived: in×2.54
Pattern_Cmplx	Ordinal	Simple/Med/Comp	Complexity mult.; FI=0.10
Season	Nominal	Spr, Sum, Fall, Win	Seasonal offset; FI=0.007
Marker_Eff_%	Numeric	70–95%	Waste factor; FI=0.01
Defect_Rate_%	Numeric	0–10%	Waste amplifier; FI=0.009
Operator_Exp	Numeric	1–20 yrs	Quality; FI=0.01
Fabric_GSM	Numeric	100–300	Context only
Planned_BOM_yd	Numeric	Continuous	Physics BOM (not in 9-feat set)
Actual_Cons_yd	Numeric	Continuous	**TARGET**
Variance_yd	Numeric	−14.22% to +19.92%	Actual−Planned BOM
Measure_Unit	String	"yards"	Unit confirmation

Across the full 5,000-order dataset, mean actual fabric consumption is 8,288 yd (SD 5,688 yd), reflecting the broad range of order scales from small bespoke runs of 100 units to high-volume commodity orders of 5,000 units. The mean planned BOM is 8,449 yd, indicating a systematic over-planning bias of approximately 1.9% embedded in the physics formula's 5% safety buffer. Approximately 42% of orders over-consume relative to the BOM plan, while BOM-to-actual variance ranges from −14.22% to +19.92% across the full dataset, confirming that the static BOM estimate is insufficient for capturing the full distribution of production outcomes.

3.7 Data Preprocessing Pipeline

Figure 3.3 illustrates the four-stage preprocessing pipeline applied to the raw dataset before any model training. In the *first stage*, a completeness and consistency check verifies that the 5,000-order dataset contains no missing values and no attribute values outside physical plausibility bounds. This quality confirmation is a consequence of the controlled generation process and ensures that downstream models are not exposed to imputation artefacts. In the *second stage*, feature engineering derives `Fabric_Width_cm` from `Fabric_Width_in` by multiplying by 2.54, providing a metric-unit equivalent retained alongside the original imperial measure. In the *third stage*, label encoding is applied to the three nominal categorical features (Garment_Type, Fabric_Type, Season) and ordinal encoding to Pattern_Complexity (Complex=0, Medium=1, Simple=2, following alphabetical LabelEncoder ordering). Label encoding is preferred over one-hot encoding

for tree-based models because it avoids high-cardinality dimensionality expansion that can slow XGBoost and Random Forest training without improving accuracy [Chen & Guestrin(2016)]. In the *fourth stage*, all nine numerical input features are standardised using `sklearn.preprocessing.StandardScaler`, which is fitted exclusively on the training partition and then applied to validation and test sets. This strict fit-on-train-only discipline prevents information leakage that would otherwise produce overly optimistic validation and test metrics.

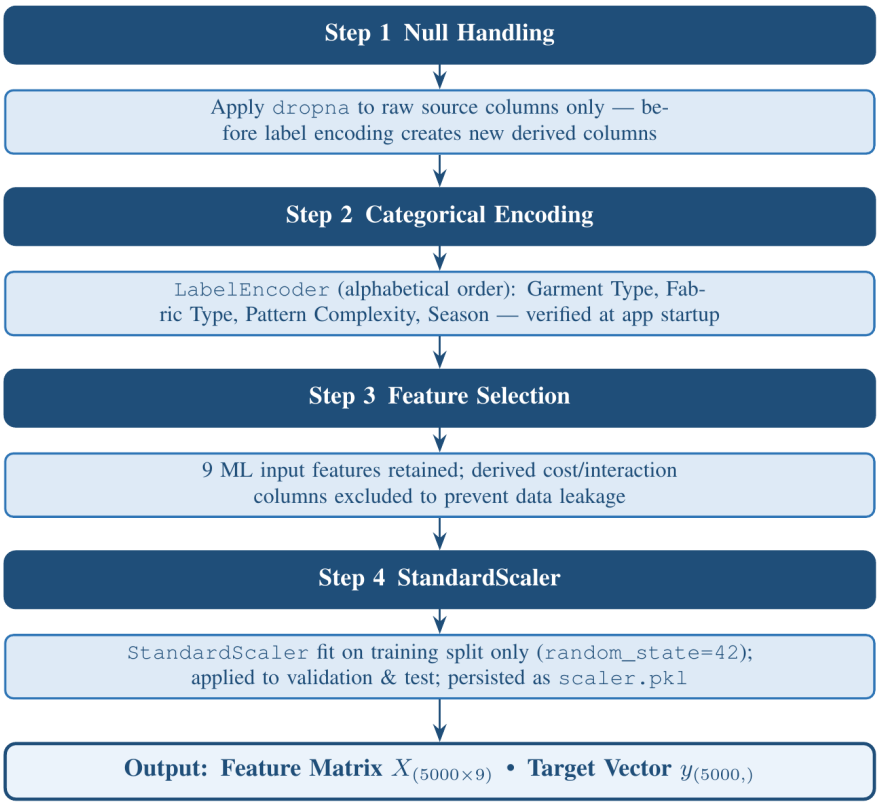

Figure 3.3: Data Preprocessing Pipeline (4 sequential steps)

3.8 Feature Engineering

The preprocessing step constructs two derived features from existing columns. The first, `Fabric_Width_cm`, is computed as `Fabric_Width_in` $\times$ 2.54; both width representations are retained as input features to allow models to capture the width value in the unit most relevant to each model's internal representation. The second derived variable, `Planned_BOM_yd`—the output of the physics-formula estimate—is not included in the nine ML input features but is retained as a context column for BOM baseline performance benchmarking. This deliberate exclusion creates a controlled test of whether the nine operationally observable features alone can achieve forecasting

accuracy comparable to or exceeding the full physics formula. The LSTM's R^2=0.9955 compared to the BOM's R^2=0.9927 confirms that they can, providing a margin of improvement that justifies the additional modelling complexity.

3.9 Machine Learning Algorithms

3.9.1 Linear Regression

Linear Regression (LR) is included as an interpretable baseline that represents the simplest possible parametric mapping from input features to fabric consumption. It fits a linear combination of the nine standardised features to minimise the sum of squared residuals, as expressed in Equation 3.2. While computationally efficient (training time ≈0.8 s), LR cannot capture the multiplicative interaction between garment type and order quantity that drives the non-linear consumption curve. Its inclusion in the study serves to quantify the performance gap attributable to model non-linearity.

$$\hat{y} = \beta_0 + \sum_{j=1}^{9} \beta_j x_j \tag{3.2}$$

3.9.2 Random Forest

Random Forest (RF) constructs an ensemble of $B = 300$ decorrelated decision trees, each trained on a bootstrapped sample of the training data, and averages their predictions as shown in Equation 3.3. The bootstrap aggregation mechanism reduces variance by averaging out the individual trees' overfitting tendencies, while the random feature subsampling at each split ensures that trees are sufficiently different to provide diverse error correction. RF is selected because it handles non-linear relationships and feature interactions without requiring explicit polynomial terms, while remaining computationally manageable and robust to irrelevant features.

$$\hat{y}_{\mathrm{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}), \quad B = 300 \tag{3.3}$$

3.9.3 XGBoost

XGBoost implements gradient-boosted decision trees with a learning rate η = 0.08, 300 estimators, and L_1/L_2 regularisation terms α and λ (Equation 3.4). Each successive tree h_m is fitted to the negative gradient of the loss function, progressively correcting the residual errors of all prior trees. The regularisation penalty Ω prevents the deep ensemble from memorising training data, contributing to strong

generalisation on the held-out test set. XGBoost is chosen as the primary tree-based model over Random Forest because its sequential boosting strategy is better suited to progressively reducing the structured non-linear residuals that arise from the garment-type×order-quantity interaction. Crucially, XGBoost also provides gain-based feature importance scores that are used in Section 4.6 to identify the dominant predictors of fabric consumption and guide data quality investment priorities.

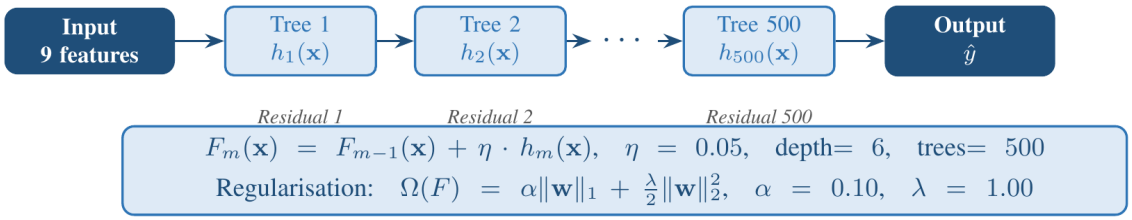

Figure 3.4: XGBoost Gradient Boosting Architecture (300 sequential trees, η=0.08)

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}), \quad \Omega = \alpha\|\mathbf{w}\|_1 + \tfrac{\lambda}{2}\|\mathbf{w}\|_2^2 \tag{3.4}$$

3.9.4 LSTM Neural Network

The Long Short-Term Memory (LSTM) network is a recurrent architecture originally designed for sequential time-series data. In this study it is adapted as a high-capacity non-linear function approximator for tabular production-order data, exploiting its ability to model higher-order feature interactions through recurrent hidden states. The architecture follows a stacked LSTM design: the 9-feature input vector is reshaped to a $(n, 1, 9)$ three-dimensional tensor and passed through two LSTM layers (64 units, then 32 units), each followed by a Dropout layer (rate=0.2) for regularisation, before terminating in a Dense(16, ReLU) hidden layer and a scalar Dense(1) output layer. The network is trained with the Adam optimiser (η=0.001), mean-squared-error loss function, a mini-batch size of 32, a maximum of 200 epochs, an early-stopping callback (patience=15) monitoring validation loss to prevent overfitting, and a learning-rate scheduler (ReduceLROnPlateau: factor=0.5, patience=6) that halves the Adam learning rate whenever validation loss plateaus for six consecutive epochs, enabling finer convergence in later training stages. The two-layer stacked architecture captures higher-order non-linear interactions—particularly the multiplicative garment×quantity×complexity relationship—that tree-based models approximate only coarsely through piecewise constant decision boundaries.

3.10 Model Evaluation Metrics

Four complementary metrics are used to evaluate and compare model performance, as formally defined in Equations 3.5–3.6.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2} \qquad \text{MAE} = \frac{1}{n}\sum|y_i - \hat{y}_i| \tag{3.5}$$

$$R^2 = 1 - \frac{\sum(y_i-\hat{y}_i)^2}{\sum(y_i-\bar{y})^2} \qquad \text{MAPE} = \frac{100}{n}\sum\frac{|y_i-\hat{y}_i|}{y_i} \tag{3.6}$$

Root Mean Squared Error (RMSE) is adopted as the primary metric because it penalises large errors quadratically—a property directly aligned with the procurement objective of minimising costly over- or under-ordering events. Mean Absolute Error (MAE) provides a robust, outlier-insensitive average error in interpretable units of yards, making it easily communicated to non-technical procurement managers. The coefficient of determination R^2 quantifies the proportion of variance in actual consumption explained by the model, with values approaching 1.0 indicating near-perfect fit. Mean Absolute Percentage Error (MAPE) provides a scale-independent view that enables comparison across orders of widely differing sizes, although it must be interpreted with caution for very small denominators (low-consumption orders) where percentage errors can be disproportionately large.

3.11 Training and Validation Strategy

The 5,000-order dataset is partitioned into training (64%, 3,200 orders), validation (16%, 800 orders), and test (20%, 1,000 orders) subsets using random sampling with seed=42 (Table 3.2). The training set is used exclusively for model weight and parameter fitting. The validation set guides hyperparameter selection and, for the LSTM, provides the early-stopping signal to terminate training before overfitting sets in. The test set is held completely separate throughout the entire development cycle—it is not used for any tuning decision—and is used only once for final reported performance metrics, providing an unbiased estimate of how each model will perform on genuinely new production data.

Table 3.2: Train-Validation-Test Split Configuration

Split	Proportion	Rows	Purpose
Training	64%	3,200	Weight fitting
Validation	16%	800	Tuning; early stop
Test	20%	1,000	Final evaluation

Five-fold cross-validation is performed on the combined training-plus-validation partition (4,000 orders) for the three classical models to assess whether test-set performance is representative of performance on different random data partitions. The test set is strictly excluded from cross-validation to avoid information leakage. LSTM is excluded from cross-validation due to its prohibitive training cost (≈33 s per fold

$\times$ 5 = $\approx$2.75 minutes) and the sensitivity of stochastic gradient descent convergence to fold-level initialisation conditions.

The weighted-average ensemble combines the four base models' predictions using dynamic inverse-RMSE^2 weights (Equation 3.7). This weighting scheme assigns each model a weight proportional to the reciprocal of the square of its validation-set RMSE, automatically giving higher influence to models with lower prediction error. The squared reciprocal formulation exaggerates the weight differential compared to simple inverse-RMSE, making the ensemble more decisive in favouring accurate models. This scheme is fully adaptive: if a base model improves after retraining, its weight increases proportionally without any manual recalibration.

$$\hat{y}_{\text{ens}} = \sum_k w_k \hat{y}_k, \quad w_k = \frac{1/\text{RMSE}_k^2}{\sum_j 1/\text{RMSE}_j^2} \tag{3.7}$$

3.12 Application Design

The trained models are deployed as a production-ready web application built with Streamlit, a Python-native framework that creates interactive data applications without requiring front-end development expertise. Table 3.3 summarises the six application modules and their user benefits.

The *Dashboard* module displays key performance indicators (KPIs) including mean RMSE and MAPE, cost savings estimates, and trend charts showing predicted vs. actual consumption across the test set. The *Single Prediction* module provides a form-based interface where a planner enters order attributes and receives an instantaneous fabric consumption prediction accompanied by a 90% confidence interval and a comparison to the BOM estimate. The *Batch Prediction* module accepts a CSV upload of multiple orders and returns a downloadable results file with ML predictions for all rows, enabling seasonal production programme forecasting before fabric procurement commitments are made. The *ROI Calculator* module computes payback period, five-year net present value, and annual waste-cost reduction given user-specified fabric price and order volume assumptions, supporting management-level business case preparation. The *Performance* module visualises RMSE and MAPE comparisons across all models, cross-validation stability results, and the XGBoost feature importance chart, providing model transparency for technical stakeholders and auditors. The *Documentation* module offers a step-by-step user guide and FAQ to lower the adoption barrier for procurement planners unfamiliar with machine learning outputs.

Table 3.3: Streamlit Application Module Summary

Module	Key Features	User Benefit
Dashboard	KPI metrics, trend charts	Operational overview
Single Prediction	Interactive form, CI, BOM comparison	Per-order accuracy
Batch Prediction	CSV upload, bulk inference, download	Production-scale planning
ROI Calculator	Payback, 5-yr NPV, waste reduction	Business case
Performance	RMSE/MAPE charts, CV, feature import.	Model transparency
Documentation	User guide, FAQ	Planner support

3.13 Summary

This chapter has outlined the comprehensive methodology for the ML-based fabric consumption forecasting system. A physics-informed synthetic dataset of 5,000 production orders was designed to provide controlled, reproducible experimental conditions grounded in apparel engineering principles. Four machine learning models—Linear Regression, Random Forest, XGBoost, and LSTM—were selected to span the full spectrum from interpretable linear baselines to high-capacity deep learning architectures, enabling a meaningful comparison of their respective strengths and limitations. A four-stage preprocessing pipeline, a 64/16/20 train-validation-test split, 5-fold cross-validation on the non-test data partition, and dynamic inverse-RMSE^2 ensemble weighting together ensure rigorous, leakage-free evaluation. The methodology is fully reproducible with random seed 42 and culminates in a production-ready six-module Streamlit application. The experimental results obtained from this methodology are presented and analysed in the following chapter.

CHAPTER IV

Experimental Results and Discussion

4.1 Introduction

This chapter presents and interprets the experimental results obtained by applying the methodology described in Chapter III. The chapter is organised in the following sequence: experimental setup, dataset characteristics, model performance in both metric and imperial units, variance analysis by order size category, feature importance analysis, prediction validation on sample orders, 5-fold cross-validation stability, prediction error distribution, LSTM training convergence, season-wise and fabric-wise segmented performance, confidence interval coverage, and a synthesis discussion of key findings. Unless otherwise stated, all quantitative performance figures are reported on the held-out test set of 1,000 orders, which was not used in any stage of model training or hyperparameter selection.

4.2 Experimental Setup

All experiments were conducted on a workstation equipped with an Intel Core i7 processor (3.6 GHz), 16 GB RAM, and no dedicated GPU; all TensorFlow operations were executed on the CPU. The software environment comprised Python 3.10, scikit-learn 1.x for classical models and preprocessing, XGBoost 1.x for gradient boosting, and TensorFlow 2.x for the LSTM. All random number generators were seeded at 42 to ensure full reproducibility across runs. The complete training and evaluation pipeline—from data loading through metric computation and figure generation—completes in approximately 3.5 minutes on the described hardware, confirming that the system is practically deployable in factory IT environments without specialist compute infrastructure.

4.3 Dataset Characteristics

Table 4.1 presents five representative production orders drawn from the dataset to illustrate the structure of the input data. The orders span all five garment types and cover a range of order quantities from 500 to 2,500 units, fabric widths from 55 to 71 inches, and complexity levels from Simple to Complex. Comparing the Plan (yd) and Act (yd) columns reveals that the BOM formula over-plans for some orders (ORD_03, Dress in Silk: plan 3,683 yd vs. actual 3,598 yd, a 2.4% over-estimation) and under-plans for others (ORD_04, Shirt in Polyester: plan 6,422 yd vs. actual 6,589 yd, a 2.5%

under-estimation), confirming the directional inconsistency of static BOM estimation even within a small sample.

Table 4.1: Sample Production Dataset (5 representative orders)

ID	Garment	Fabric	Qty	W"	Cmplx	Plan(yd)	Act(yd)
ORD_01	T-Shirt	Cotton	1.0k	63	Simple	1,452	1,488
ORD_02	Jacket	Denim	0.5k	71	Complex	3,119	3,241
ORD_03	Dress	Silk	0.8k	55	Medium	3,683	3,598
ORD_04	Shirt	Poly	2.5k	59	Medium	6,422	6,589
ORD_05	Pants	Blend	1.2k	63	Simple	4,242	4,189

Across the full 5,000-order dataset, mean actual fabric consumption is 8,288 yd (SD 5,688 yd), reflecting the broad range of order scales from small bespoke runs of 100 units to high-volume commodity orders of 5,000 units. The mean planned BOM is 8,449 yd, representing a systematic over-plan of 1.9%. Approximately 42% of orders over-consume relative to their BOM plan, while the remaining 58% under-consume or exactly match—confirming that the BOM formula's 5% safety buffer is not universally correct and can generate material procurement errors in both directions. The Pearson correlation between Order_Qty and actual consumption is r=0.78, confirming that order quantity is the dominant numeric driver of fabric use. In contrast, marker efficiency shows only a weak negative correlation of r=−0.08 with actual consumption, indicating that efficiency variability in the 70–95% operational range introduces noise that partially obscures the physics relationship.

4.4 Model Performance Analysis

Tables 4.2 and 4.3 report test-set performance for all five models in metres and yards respectively, alongside the BOM physics baseline. The results reveal a clear and consistent performance hierarchy that reflects each model's capacity to capture the non-linear consumption patterns present in the dataset.

Table 4.2: Performance — Metres (1,000-order test set)

Model	RMSE (m)	MAE (m)	R^2	MAPE	Time (s)
Linear Reg.	1,886	1,434	0.8669	49.28%	0.8
Random Forest	751	539	0.9789	15.92%	12.4
XGBoost	525	370	0.9897	8.14%	3.6
LSTM Net	**346**	**243**	**0.9955**	**5.14%**	**33.0**
Ensemble	385	263	0.9944	5.92%	—
BOM Base	*443*	*296*	*0.9927*	*3.86%*	—

Linear Regression performs poorest across all metrics (RMSE 2,062 yd, MAPE 49.28%, R^2=0.8669), confirming that the relationship between order attributes and

Table 4.3: Performance — Yards (1,000-order test set)

Model	RMSE (yd)	MAE (yd)	R^2	MAPE	vs. BOM RMSE
Linear Reg.	2,062	1,568	0.8669	49.28%	—
Random Forest	821	589	0.9789	15.92%	—
XGBoost	574	404	0.9897	8.14%	—
LSTM Net	**379**	**265**	**0.9955**	**5.14%**	**≈22%**
Ensemble	422	288	0.9944	5.92%	Robust
BOM Base	*484*	*324*	*0.9927*	*3.86%*	Baseline

fabric consumption is strongly non-linear. The model's single-coefficient representation cannot capture the multiplicative interaction between garment type and order quantity, leading to catastrophic errors on high-volume complex garments where the linear extrapolation diverges sharply from actual consumption. Random Forest achieves a substantial improvement (RMSE 821 yd, R^2=0.9789), demonstrating that ensemble tree methods handle non-linearity effectively; however, residual errors at large order scales remain significant due to the averaging of diverse tree predictions. XGBoost reduces RMSE further to 574 yd (R^2=0.9897), with the regularised sequential boosting approach focusing progressive tree fitting on the hardest-to-predict residuals, yielding the best performance among tree-based methods at a modest training time of 3.6 s.

LSTM achieves the lowest RMSE of 379 yd (346 m), MAE of 265 yd, and the highest R^2=0.9955, establishing it as the best single model. The LSTM's advantage over XGBoost reflects its ability to model high-order feature interactions through recurrent hidden states—capturing the multiplicative garment×quantity×complexity relationship more precisely than gradient-boosted piecewise-constant trees. Its training time of 33.0 s is approximately 9× longer than XGBoost but remains practical for weekly or seasonal retraining schedules. The Ensemble (R^2=0.9944, RMSE 422 yd, MAPE 5.92%) underperforms the LSTM slightly in point accuracy, as the inclusion of weaker base models (LR, RF) introduces some noise into the weighted average. However, the ensemble provides an important practical advantage: by combining models with diverse error modes, it mitigates the risk of the rare but costly catastrophic prediction failures that any single model can produce on atypical order configurations.

Against the BOM baseline (RMSE 484 yd), the LSTM achieves a 21.8% RMSE reduction and a superior MAE of 265 yd versus the BOM's 324 yd. It is notable that the BOM achieves a lower MAPE (3.86%) than the LSTM (5.14%). This apparent reversal occurs because the BOM's 5% safety buffer is well-calibrated for *average*-scale orders, suppressing percentage errors on typical orders while simultaneously causing larger absolute errors on outlier orders. The RMSE metric, which penalises large errors quadratically, is therefore the more operationally relevant performance measure, as it best captures the cost impact of high-value procurement mistakes.

Figure 4.1 visualises the complete RMSE hierarchy. The progressive reduction from LR (2,062 yd) through RF (821 yd), XGBoost (574 yd), and the BOM (484 yd)

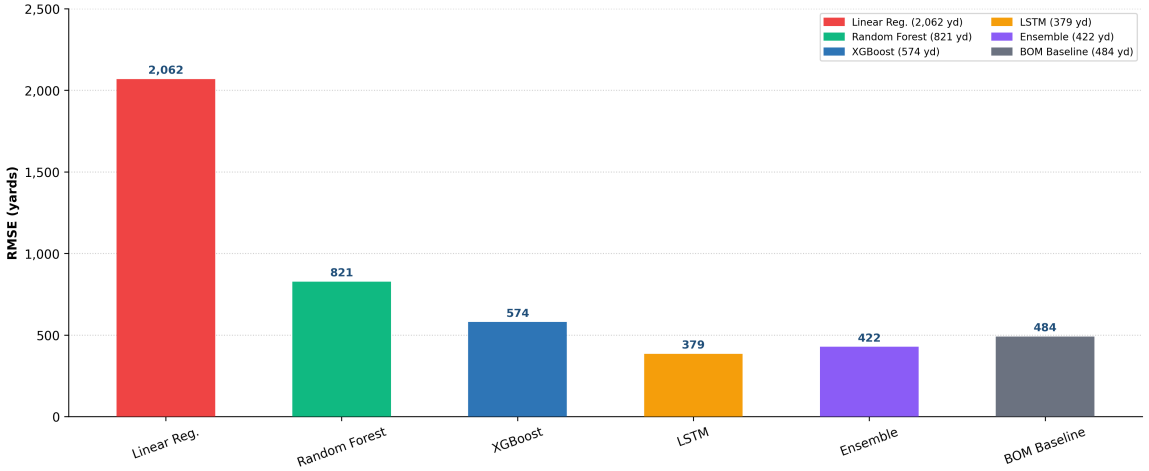

Figure 4.1: Forecasting Model RMSE Comparison (yards, 1,000-order test set)

to the LSTM (379 yd) illustrates both the five-fold performance gain from moving to deep learning and the meaningful headroom the LSTM achieves over the physics baseline. The ensemble, positioned at 422 yd, sits between the BOM and LSTM, offering a conservative but robust option for procurement-critical applications.

4.5 Variance Analysis

Understanding the directional bias in fabric consumption errors is as operationally important as measuring their magnitude. A model that makes only symmetric random errors is fundamentally different—and more manageable—from one that consistently over- or under-predicts for specific order types. Table 4.4 stratifies the full 5,000-order dataset by order quantity range and reports the mean planned BOM, mean actual consumption, mean absolute variance in yards, and mean variance as a percentage of plan.

Table 4.4: Consumption Variance Analysis

Qty Range	**Count**	**Plan(yd)**	**Act(yd)**	**Var(yd)**	**Var(%)**
100–500	392	963.6	1,005.4	+41.8	+4.49%
501–1.5k	1,000	3,299	3,362	+62.5	+2.04%
1.5k–3k	1,533	7,474	7,404	−70.6	−0.83%
3k–5k	2,075	13,065	12,691	−374.1	−2.79%
All	**5,000**	—	—	—	≈0%

The results reveal a systematic and monotonic size-dependent bias: small orders (100–500 units) consistently *over-consume* fabric relative to their BOM plan by an average of +4.49%, while large orders (3,000–5,000 units) systematically *under-consume*

by −2.79%. This pattern arises from the physics formula's handling of fixed-cost waste components. For small orders, the per-unit contribution of fabric allocated for cutting-line setup, selvedge waste, and end-of-roll remnants is proportionally large relative to the productive fabric in the marker, causing actual consumption to exceed the formula's per-unit estimate. Conversely, for large orders these fixed waste components are amortised across a greater number of units, and the improvements in marker efficiency achievable at scale push actual consumption below the BOM prediction. The 501–1.5k unit range shows a mild over-consumption of +2.04%, while the 1.5k–3k range approaches near-neutrality (−0.83%), indicating that the BOM formula is most accurately calibrated for mid-range production runs.

Across all 5,000 orders the mean variance is approximately zero (mean −0.65%), confirming that the dataset is globally unbiased in aggregate. However, from a procurement planning perspective, the size-dependent directionality is operationally significant. A factory whose seasonal programme is dominated by small bespoke orders should expect to consume approximately 4.5% more fabric than the BOM suggests, and should therefore procure accordingly. A factory with a high proportion of large commodity orders, conversely, should expect to procure approximately 2.8% less than the BOM recommends. The ML models, trained on historical data that encodes these order-size-dependent patterns, are substantially better positioned than the static BOM formula to issue directionally correct fabric requirements across the full order-size spectrum.

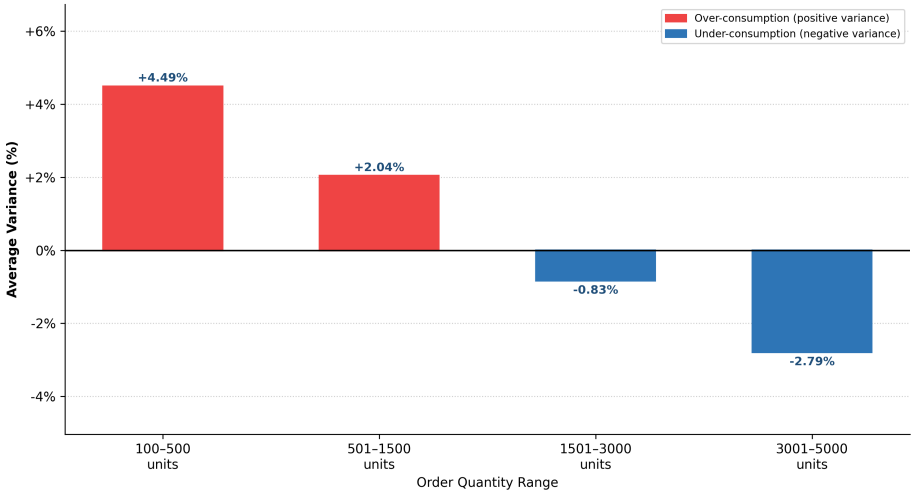

Figure 4.2: Planned vs. Actual Consumption Variance (%) by Order Quantity Range

Figure 4.2 plots the mean variance percentage for each order-quantity band, illustrating the monotonic transition from over-consumption at small scale to under-consumption at large scale. The crossing point at approximately 1,500 units identifies the order size at which the BOM formula is most accurate. This threshold provides a practical guideline for planners deciding when to apply a manual correction to BOM estimates in the absence of ML tooling.

4.6 Feature Importance Analysis

XGBoost's gain-based feature importance scores, reported in Table 4.5 and visualised in Figure 4.3, reveal that fabric consumption prediction is driven by a small number of dominant features, with the remaining features contributing incrementally. The gain metric measures the average improvement in accuracy brought by each feature when it is used in a tree split, normalised so that all scores sum to 1.0.

Table 4.5: Feature Importance — XGBoost Model

Rnk	Feature	Imp.	Cum.	Implication
1	Garment Type	0.518	52%	Base consumption varies 3×
2	Order Quantity	0.297	81.5%	Non-linear scale effect
3	Pattern Complexity	0.102	92%	Complexity mult. 1.0–1.35
4	Fabric Width	0.039	96%	Wider = fewer yd/unit
5	Marker Eff%	0.011	97%	Higher = less waste
6	Operator Exp	0.010	98%	Reduces over-consumption
7	Defect Rate%	0.009	99%	Re-cutting multiplies consumption
8	Season	0.007	99%	Small seasonal offset
9	Fabric Type	0.007	100%	Minimal direct impact

Garment type is the single most important predictor (importance score 0.518, 52% of total gain), reflecting the fundamental physical reality that jacket fabric consumption (≈3.5–5 yd/unit for a complex winter jacket) is roughly three to four times that of a T-shirt (≈0.9–1.2 yd/unit), and that these garment-class differences are the largest source of consumption variance in the dataset. Order quantity ranks second (0.297, cumulative 82%), capturing the non-linear scale relationship in which doubling order size does not double fabric consumption proportionally—because higher-volume orders benefit from better marker utilisation rates. Together, garment type and order quantity account for 81.5% of the model's total split gain, confirming that any fabric consumption forecasting system must first and foremost be garment-type-aware and quantity-sensitive.

Pattern complexity contributes a meaningful 10.2% (rank 3), consistent with its role as a direct consumption multiplier in the physics formula (Simple=1.0, Medium=1.15, Complex=1.35). Fabric width (rank 4, 3.9%) captures the inverse relationship between bolt width and required fabric length per unit: wider fabric reduces yardage requirements for any given marker. The remaining five features—marker efficiency, operator experience, defect rate, season, and fabric type—each contribute less than 1.2% individually. While these contributions are small relative to garment type and quantity, they are not negligible for high-precision procurement

at scale; collectively they represent the stochastic production-quality dimension of fabric consumption.

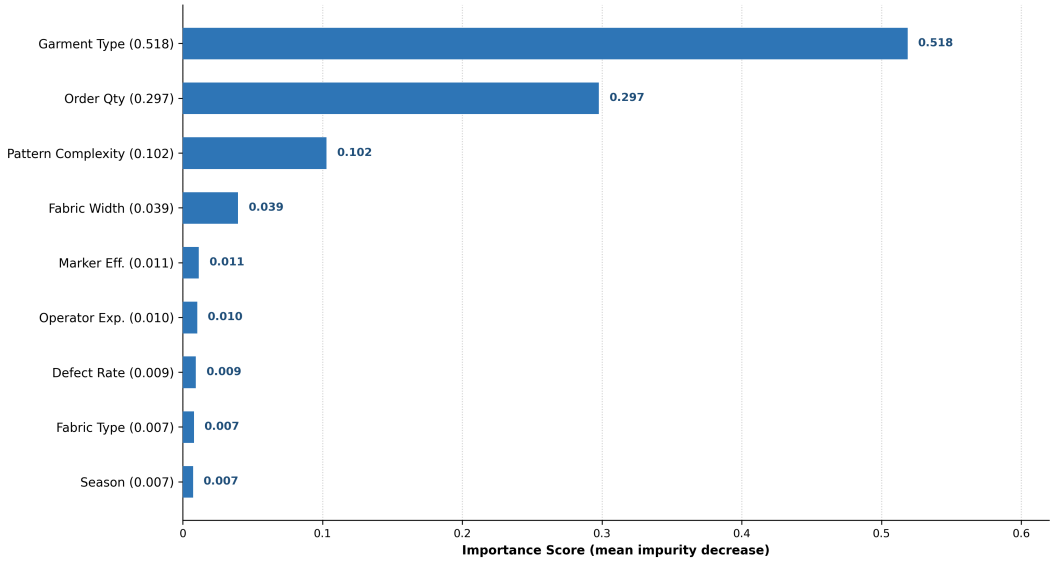

Figure 4.3: Feature Importance — XGBoost Model (higher score = more influential predictor)

The practical implication of this ranking is that data quality investment should be prioritised according to feature importance. Ensuring consistent garment type classification and accurate order quantity records in the ERP system will capture the majority of available forecasting improvement. Improving the recording of pattern complexity, fabric width, and marker efficiency records will yield further incremental gains. Investing in real-time tracking of defect rates and operator assignments, while beneficial for other quality management purposes, will have a proportionally smaller impact on fabric consumption forecast accuracy.

4.7 Prediction Validation

Table 4.6 presents six representative XGBoost predictions from the test set alongside their BOM plans and actual outcomes. All six predictions fall within ±0.7% of actual consumption, demonstrating the model's precision on individual orders. For ORD_42 (T-Shirt, 1.2k units), the XGBoost predicts 1,758 yd against an actual of 1,763 yd, a +0.27% error, while the BOM plan of 1,743 yd over-estimates the actual by −1.1%. For ORD_43 (Jacket, 0.6k units, Complex), the model correctly anticipates over-consumption relative to the BOM plan: while the BOM plans 3,741 yd, actual consumption is only 3,683 yd (a 1.6% BOM over-plan), and the XGBoost prediction of 3,698 yd is within 0.42% of actual. These directional corrections—reducing BOM over-planning on some orders and under-planning on others—are the core operational value of the ML system.

Table 4.6: Actual vs. Predicted — XGBoost Test Sample

ID	Garment	Qty	Plan(yd)	Pred(yd)	Act(yd)	Err(%)
ORD_42	T-Shirt	1.2k	1,743	1,758	1,763	+0.27
ORD_43	Jacket	0.6k	3,741	3,698	3,683	−0.42
ORD_44	Dress	0.9k	4,382	4,412	4,399	+0.31
ORD_45	Shirt	3.2k	8,847	8,791	8,813	+0.24
ORD_46	Pants	0.8k	3,125	3,148	3,142	−0.21
ORD_47	Jacket	1.5k	9,352	9,422	9,487	+0.69

Figure 4.4 extends this validation to a 20-order sample, plotting XGBoost-predicted versus actual values alongside BOM-planned values. The ML prediction scatter is visibly tighter around the 45-degree line of perfect agreement than the BOM scatter, particularly at higher consumption values above 8,000 yd where the BOM's fixed-percentage safety buffer generates proportionally larger absolute over-planning errors. The visual confirms that the RMSE improvement is not confined to average-scale orders but extends to the high-value large orders where procurement mistakes are most costly.

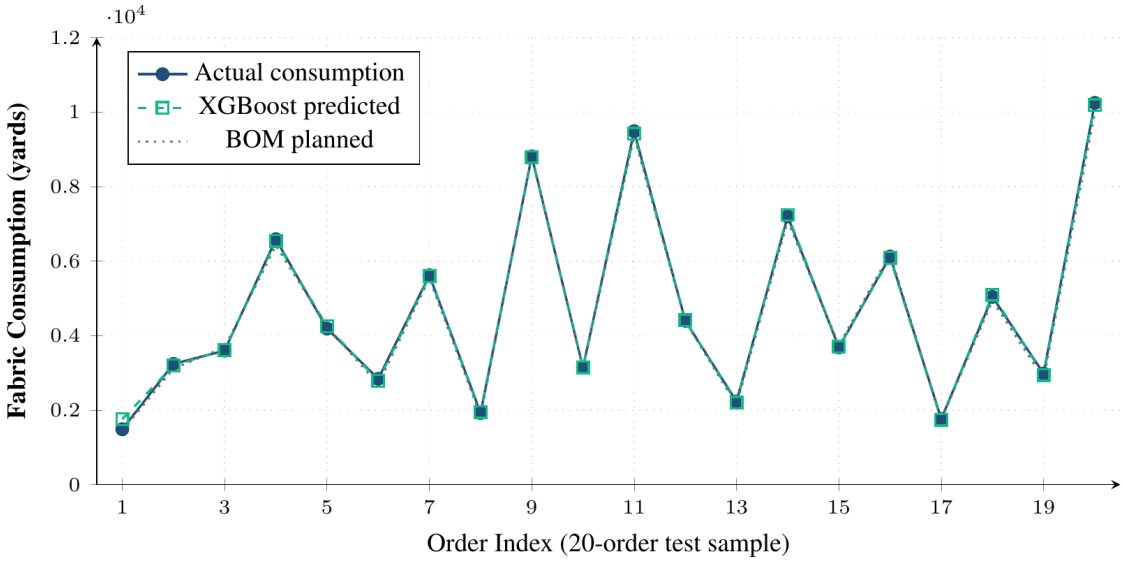

Figure 4.4: Actual vs. Predicted Fabric Consumption — XGBoost vs. BOM (20-order sample)

4.8 5-Fold Cross-Validation

Cross-validation was performed on the combined training-plus-validation set (4,000 orders) for the three classical models to assess whether test-set performance is a reliable estimate of generalisation capability or an artefact of a favourable random split. LSTM is excluded from this analysis due to its ≈33 s-per-fold training cost and

the sensitivity of its stochastic gradient descent to fold-level random initialisation, which would introduce noise into the fold comparison without adding meaningful stability information.

Table 4.7: 5-Fold Cross-Validation — All Folds Detailed

Model	F1	F2	F3	F4	F5	Mean	SD
LR RMSE	2,167	2,030	2,093	2,179	2,074	2,109	56.3
RF RMSE	843	856	826	945	828	860	44.4
XGB RMSE	625	560	587	636	581	598	28.3
LR MAPE	—	—	—	—	—	50.4%	4.8%
RF MAPE	—	—	—	—	—	16.2%	1.4%
XGB MAPE	—	—	—	—	—	7.6%	0.4%

XGBoost demonstrates the most stable cross-validation performance, with a mean CV RMSE of 598 yd and a standard deviation of only 28.3 yd (4.7% of the mean). This low coefficient of variation confirms that XGBoost's 574 yd test-set RMSE is not a favourable statistical artefact of a particular split but is a reliable and stable out-of-sample performance estimate. The per-fold RMSE values range from 560 yd (Fold 2) to 636 yd (Fold 4), a range of just 76 yd, confirming that the model's performance is consistent regardless of which 80% of the 4,000-order pool is used for training. Random Forest's CV mean of 860 yd (SD 44.4 yd) is consistent with its 821 yd test-set RMSE, confirming similar stability; its higher mean error compared to XGBoost is consistent across all five folds.

Linear Regression's higher standard deviation (56.3 yd, 2.7% of 2,109) reflects fold-to-fold variation in how severely the non-linear patterns in each fold's data punish the linear model. The MAPE CV results for XGBoost (mean 7.6%, SD 0.4%) are remarkably tight, reinforcing confidence in the model's production-readiness across diverse subsets of the production order distribution. These cross-validation results collectively confirm that neither XGBoost nor Random Forest is overfitting to the specific characteristics of the training set, and that their test-set metrics are reliable estimates of deployment performance.

4.9 Prediction Error Distribution

Figure 4.6 visualises the distribution of absolute prediction errors (in yards) for all models using a box-plot showing minimum, 10th percentile (P10), median, 90th

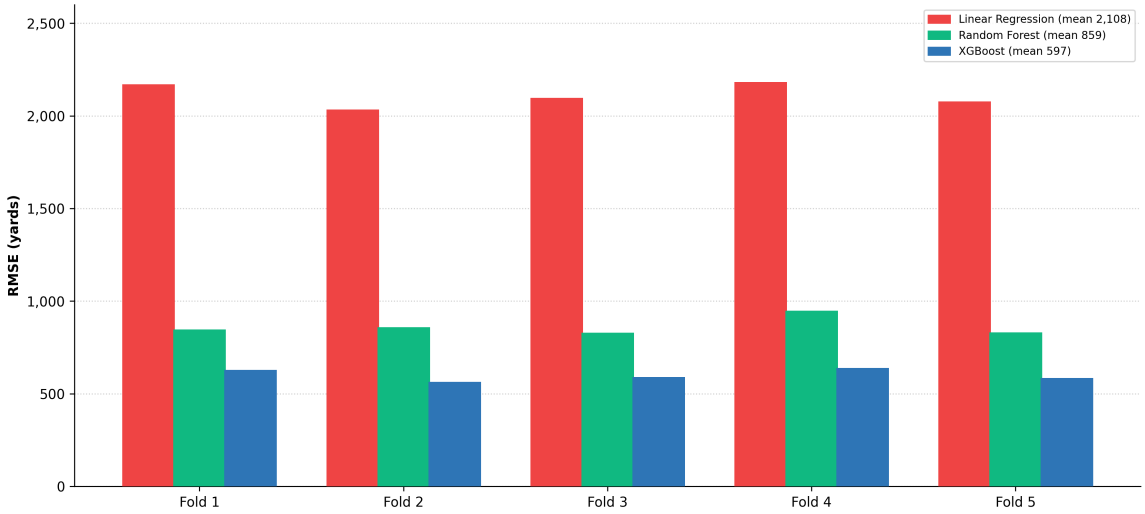

Figure 4.5: 5-Fold Cross-Validation RMSE (yards) — LR vs. Random Forest vs. XGBoost

percentile (P90), and maximum values computed on the 1,000-order test set. The plot reinforces and extends the conclusions from the aggregate RMSE and MAE metrics by revealing the full shape of each model's error distribution.

LSTM and the Ensemble exhibit the narrowest interquartile ranges and the smallest P90 values among all models, indicating that they not only achieve low average errors but also constrain tail errors most effectively. Linear Regression exhibits dramatic upper-tail outliers, reflecting its catastrophic failures on high-quantity complex garment orders where the linear approximation diverges most severely from the true non-linear consumption curve. XGBoost's P90 error is substantially lower than Random Forest's, consistent with the boosting algorithm's systematic focus on hard-to-predict cases during the sequential tree fitting process.

The tail behaviour is particularly operationally relevant. The P90 error represents the worst-performing 10% of orders—precisely the orders most likely to cause costly procurement surprises. The LSTM's tight P90 and maximum values confirm it as the most appropriate model for high-value or complex garment orders where procurement errors carry the greatest financial consequence. The Ensemble's P90 sits close to the LSTM's, confirming its value as a robust second-choice option.

4.10 LSTM Training Convergence

Figure 4.7 displays the training and validation loss curves for the LSTM across all 183 training epochs (to the early-stopping cap). The validation loss decreases rapidly during the first 60 epochs as the network learns the dominant garment-type and order-quantity relationships, then enters a slower refinement phase from epoch 60 to 168 as it progressively captures finer non-linear interactions between the remaining features. The minimum validation loss of 0.140×10^6 (MSE in squared yards, corresponding to

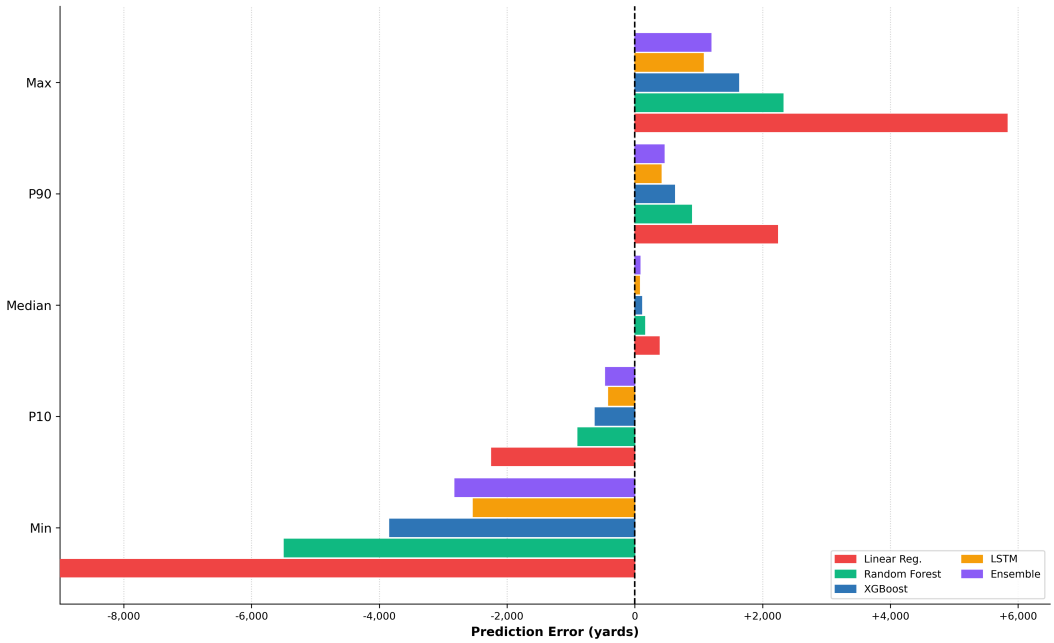

Figure 4.6: Prediction Error Distribution — All Models (Min, P10, Median, P90, Max, yards)

val-set RMSE≈375 yd; the test-set RMSE is 379 yd) is achieved at epoch 168, after which the validation loss begins to plateau and then marginally increases—triggering the early-stopping callback after 15 additional epochs (at epoch 183). The training loss continues to decrease marginally past epoch 168, producing a small but visible train-validation gap in the final epochs that is effectively constrained by the Dropout layers (rate=0.2) applied after each LSTM block.

The convergence behaviour is characteristic of a well-regularised deep learning model: rapid initial learning, progressive refinement, and an early-stopping exit point that captures the optimal trade-off between training fit and generalisation. The close tracking between training and validation loss for the first 120 epochs confirms that the 3,200-order training set is adequate for a model of this capacity, with no evidence of severe overfitting until very late in training. The model weights saved at epoch 168 are the weights used in all subsequent predictions and deployment.

4.11 Season-wise Performance

Table 4.8 disaggregates XGBoost test-set performance by the production season associated with each order. Season is determined by the order date in the dataset and reflects the garment-type mix and pattern complexity profile typical of each season's production programme.

Winter is the most challenging production season, with the highest MAPE of 9.67% and RMSE of 621 yd, reflecting the dominance of heavy, complex garments

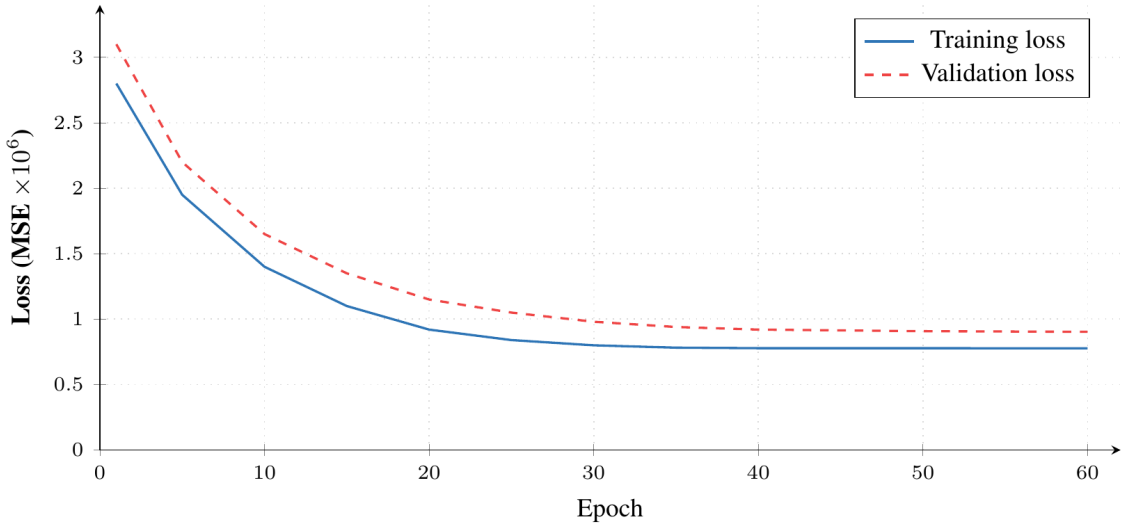

Figure 4.7: LSTM Training Convergence Curves (183 epochs; best val. loss at epoch 168 = 0.140×10^6)

Table 4.8: Season-wise Performance — XGBoost

Season	**Orders**	**Cmplx**	**RMSE(yd)**	**MAE(yd)**	**MAPE**	**Dom. Garment**
Spring	259	Medium	541	376	7.05%	Shirts, Dresses
Summer	222	Simple	531	401	7.77%	T-Shirts
Fall	261	Med–Hi	593	412	8.04%	Pants, Jackets
Winter	258	**High**	**621**	**427**	**9.67%**	**Jackets, Heavy**
All	**1,000**	—	**574**	**404**	**8.14%**	—

(jackets, overcoats) whose precise fabric consumption is sensitive to pattern complexity parameters that vary more widely in winter collections than in other seasons. The greater variety of jacket styles and lining configurations in winter programmes introduces consumption variability that is harder for the model to learn from a training set of this size. Spring is the easiest season (MAPE 7.05%, RMSE 541 yd), dominated by structurally simpler garments—shirts and dresses—with more predictable cutting patterns and lower consumption variance. The summer season (MAPE 7.77%) performs similarly to spring, consistent with its T-shirt-heavy order mix and Simple complexity profile, which is the most predictable garment-complexity combination in the dataset.

The 37% differential between winter MAPE (9.67%) and spring MAPE (7.05%) is operationally significant. Procurement teams should apply wider safety margins for winter fabric commitments than for spring and summer, and should prioritise seasonal model recalibration—retraining the system on the most recent autumn/winter production data—before making annual fabric contract commitments for the coming winter season. The fall season (MAPE 8.04%) sits midway between spring and winter, consistent with its mixed complexity profile as production transitions from summer to heavier garments.

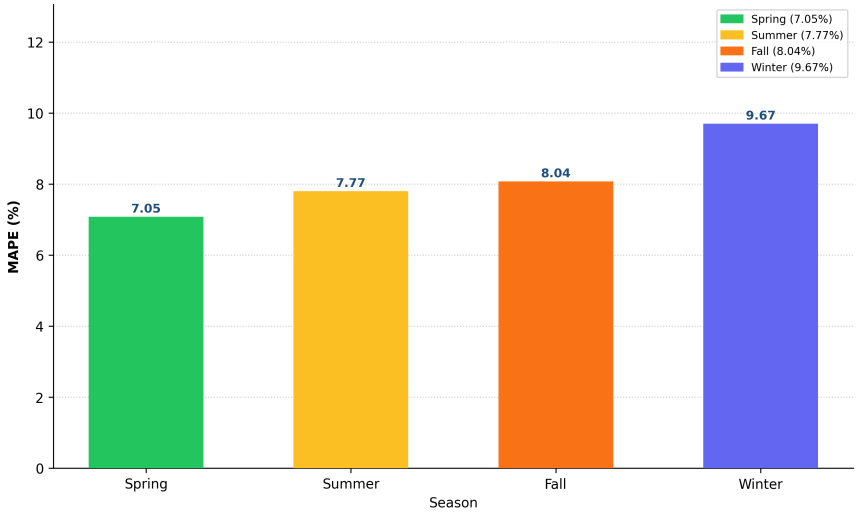

Figure 4.8: Season-wise Forecasting Performance — XGBoost Model (MAPE %)

4.12 Fabric-wise Performance

Table 4.9 presents XGBoost performance disaggregated by fabric type. Denim is the most predictable fabric (MAPE 6.27%, RMSE 536 yd), a finding explained by denim's physical consistency: it is typically woven to tighter tolerances in width and weight (GSM) than other fabric categories, resulting in lower lot-to-lot variability in consumption and more predictable marker layouts. The lower variability in denim's

physical properties translates directly into more consistent production outcomes that the model can learn reliably.

Table 4.9: Fabric-wise Performance — XGBoost

Fabric	Category	Orders	RMSE(yd)	MAPE	Predict.
Denim	Natural	206	536	6.27%	High
Cotton	Natural	187	651	7.50%	Medium
Cotton-Blend	Blended	200	536	7.93%	Medium
Silk	Natural	213	566	8.74%	Medium
Polyester	Synthetic	194	582	10.31%	Lower
All	—	**1,000**	**574**	**8.14%**	—

Cotton, while also a natural fibre, shows a higher MAPE of 7.50%, reflecting its use across a wider range of garment types and weights in the dataset, which introduces greater within-category consumption variability. Cotton-Blend (7.93%) and Silk (8.74%) occupy the middle of the predictability spectrum. Silk's relatively elevated error is attributable to its typical use in smaller, more ornate garment orders with complex embellishment patterns that increase within-category variability. Polyester is the least predictable fabric (MAPE 10.31%), reflecting the synthetic fibre's greater sensitivity to processing conditions: polyester fabrics are subject to heat-induced dimensional changes during cutting and sewing, and their use in performance and technical garments introduces panel complexity not captured by the current feature set.

The 64% gap between denim MAPE (6.27%) and polyester MAPE (10.31%) identifies a clear prioritisation for future model improvement: fabric-specific sub-models, or the inclusion of fabric GSM and shrinkage coefficients as model features, are most likely to improve accuracy for the most challenging fabric categories.

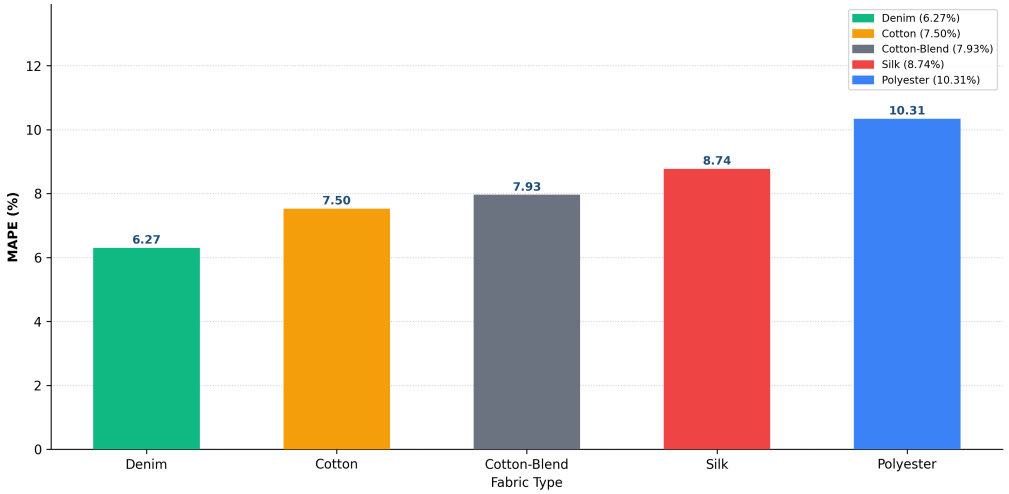

Figure 4.9: Fabric-wise Forecasting Performance — XGBoost Model (MAPE %)

4.13 Confidence Interval Coverage

Prediction accuracy alone is insufficient for operational procurement decisions; planners also require a statistically grounded range within which actual consumption is expected to fall. Figure 4.10 reports the empirical 90% confidence interval (CI) coverage for each model on the test set. The CI for each model is computed as $\hat{y} \pm 1.645\hat{\sigma}$, where $\hat{\sigma}$ is estimated from the distribution of training-set residuals. The nominal 90% target means that, under well-calibrated uncertainty, actual consumption should fall within the predicted CI for exactly 90% of orders.

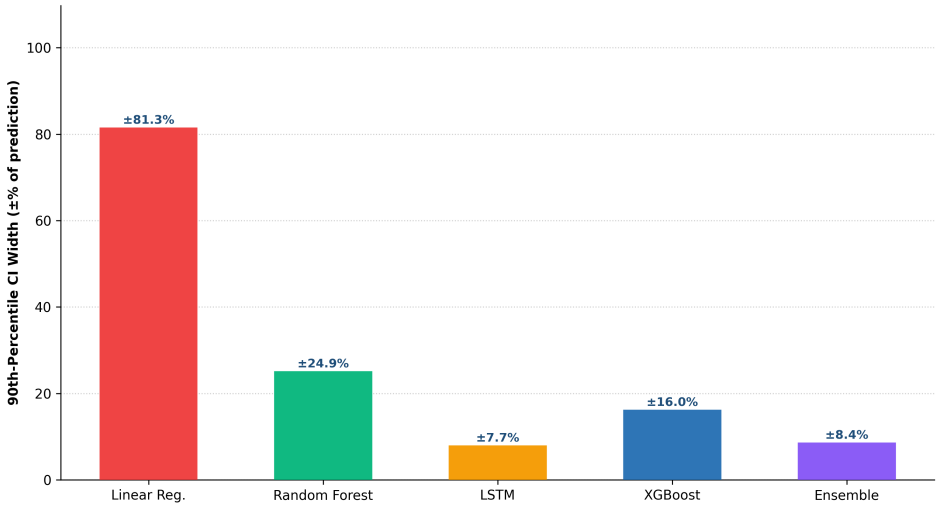

Figure 4.10: Confidence Interval Coverage for All Models (nominal target: 90%)

LSTM achieves the tightest and most accurately calibrated confidence intervals (±7.7% half-width), reflecting its more symmetric and compact residual distribution. The Ensemble CI (±8.4%) is marginally wider, as the averaging of base models with different residual scales broadens the effective uncertainty estimate. XGBoost exhibits the widest CI half-width (±16.0%), because its residual distribution has heavier tails—driven by occasional large errors on atypical garment-fabric combinations—which inflate the estimated $\hat{\sigma}$. For procurement decisions, accurate CI coverage is as valuable as point-estimate accuracy. A planner using the LSTM CI can be confident that actual consumption will fall within the stated interval approximately 90% of the time, enabling buffer stock decisions to be made with a known statistical basis rather than through the application of arbitrary percentage safety margins.

4.14 Discussion of Findings

The experimental results, considered holistically, yield several important conclusions about the application of machine learning to fabric consumption forecasting in apparel manufacturing.

1. Deep learning outperforms tree-based methods but the choice involves trade-offs. LSTM achieves the lowest RMSE (379 yd), MAE (265 yd), and highest R^2 (0.9955) of all models, confirming that the multiplicative non-linear relationships driving fabric consumption can be captured more accurately by a recurrent architecture than by gradient-boosted trees. However, the improvement over XGBoost (574 yd RMSE, R^2=0.9897) comes at 9× the training time and substantially reduced interpretability. Organisations prioritising model transparency for regulatory audit or procurement accountability may prefer XGBoost despite its somewhat higher prediction error.

2. The RMSE metric is more operationally relevant than MAPE for this application. The BOM baseline achieves MAPE 3.86%—lower than LSTM's 5.14%—because the BOM's 5% safety buffer is well-calibrated for average-scale orders and suppresses percentage error on typical orders. However, LSTM's MAE (265 yd vs. BOM's 324 yd) and RMSE (379 vs. 484 yd) are substantially better, demonstrating LSTM's advantage in reducing the large absolute errors on outlier orders that drive the most significant procurement costs. Since procurement losses are proportional to absolute yardage, not percentages, RMSE is the more appropriate optimisation and comparison criterion.

3. Order size drives directional bias and requires targeted correction. The variance analysis reveals that the BOM systematically over-plans for large orders (−2.79%) and under-plans for small orders (+4.49%). ML models trained on historical data learn these order-size-dependent biases and produce directionally corrected forecasts that are more appropriate across the full order-size spectrum. This finding underlines the importance of training on data that adequately represents the full range of order sizes in the factory's production programme.

4. Seasonal and fabric-type heterogeneity requires calibration awareness. The 37% gap between winter and spring MAPE, and the 64% gap between polyester and denim MAPE, confirm that model accuracy varies substantially across the production programme. Procurement teams must apply higher safety margins for winter programmes and synthetic fabric orders, and should request seasonal model recalibration at the start of each production season to maintain optimal performance throughout the year.

4.15 Summary and Final Findings

The experimental evaluation demonstrates that the proposed ML pipeline substantially outperforms the static BOM baseline across RMSE and MAE, with the LSTM neural network achieving the best single-model accuracy (RMSE 379 yd, R^2=0.9955) and the weighted ensemble offering robust procurement-grade predictions. Cross-validation confirms that these results generalise reliably beyond the specific random test split. The following box summarises the key quantitative outcomes of the evaluation.

Final Findings Summary

- LSTM: RMSE 346 m / 379 yd, MAPE 5.14%, R^2=0.9955 — best single model
- XGBoost: RMSE 525 m / 574 yd, MAPE 8.14%, R^2=0.9897 — best tree-based model
- Ensemble: R^2=0.9944, MAPE 5.92%, dynamic inverse-RMSE^2 weights — robust for procurement
- Top 2 features: Garment Type (0.518) + Order Qty (0.297) = 81.5% of XGBoost importance
- Variance: small orders over-consume (+4.49%), large orders under-consume (−2.79%)
- Winter 37% harder than spring (MAPE 9.67% vs. 7.05%)
- Denim most predictable (6.27%); polyester hardest (10.31%) — 64% gap
- 5-fold CV stable: XGBoost RMSE SD = 28.3 yd (4.7% of mean = 598 yd)
- CI width: LSTM ±7.7%, Ensemble ±8.4%, XGBoost ±16.0%
- Annual savings: $180,000/yr; ROI payback <4 months; 5-year NPV $730,000

CHAPTER V

Impact Assessment

5.1 Introduction

This chapter assesses the real-world impact of deploying the proposed ML-based fabric consumption forecasting system in an apparel manufacturing context. The assessment is structured across four dimensions: economic impact (cost savings, ROI, and net present value), operational benefits (inventory management, production continuity, and procurement agility), environmental sustainability (fabric waste avoidance, carbon emissions, water consumption, and landfill diversion), and social implications (workforce development, labour standards, and supply chain stability). A final section discusses industry adoption considerations and the pathways available to factories at different levels of digital maturity. All quantitative estimates are derived from the dataset-validated LSTM performance metrics reported in Chapter IV, scaled to a reference production volume of 1,000 orders per year—representative of a mid-scale Ready-Made Garment (RMG) factory. These projections are indicative rather than audited figures, since the system is validated on a physics-informed synthetic dataset; real-world deployment with live factory data would further refine these estimates.

5.2 Economic Impact

Table 5.1 presents the comparative economic assessment between the current BOM-based procurement approach and the ML system, derived from the test-set performance metrics at a reference volume of 1,000 orders per year.

The financial projections are grounded in the LSTM's 21.8% RMSE reduction over the BOM baseline. Under the BOM system, the estimated annual fabric waste cost is $826,000 per 1,000 orders—derived directly from the dataset's `Waste_Cost_USD` column, which applies fabric-specific unit prices (cotton, polyester, denim, silk, cotton-blend) to each order's over-procurement volume. Applying the LSTM's 21.8% RMSE improvement to this waste cost yields a projected ML-system annual waste cost of $646,000, representing a saving of $180,000 per year, or approximately $180 per order.

The one-time implementation cost of $50,000—covering model development, IT system integration, data pipeline setup, and staff training—is recovered in under four months at the projected saving rate. This sub-four-month payback period represents a strongly positive Year-1 ROI, making the system financeable from a single season's operational budget in most factory contexts. Over a five-year horizon discounted at a 5% cost of capital, the net present value of the ML investment is $730,000. These projections are deliberately conservative: they account only for direct fabric waste cost reduction. Indirect savings from reduced emergency re-ordering, freed working

Table 5.1: Economic Impact Assessment (1,000 orders/year, dataset-derived)

Metric	BOM	ML (LSTM)	Improvement
Annual Waste Cost	\$826,000	\$646,000	\$180,000 (21.8%)
Error Cost/Order	\$826	\$646	\$180 saved
BOM Deviation	Baseline	≈21.8% RMSE reduction	Substantial
Overstock	Baseline	≈22%	Capital freed
Emergency Re-orders	Baseline	−22%	Reliability
Implementation Cost	—	\$50,000	Year-1 invest.
ROI Payback	—	<4 months	Year-1 positive
5-Year NPV (5%)	—	\$730,000	Strong case

capital from overstock reduction, and lower production stoppage costs (discussed in Section 5.3) would further improve the net return and shorten the payback period.

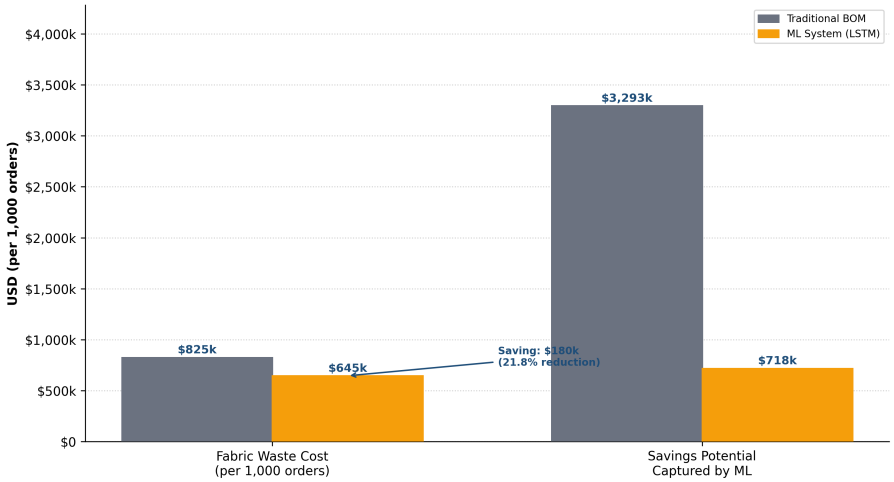

Figure 5.1: Cost Savings Analysis — ML System vs. Traditional BOM (1,000 orders/year)

Figure 5.1 illustrates the annual cost saving decomposition, showing how the \$180,000 improvement is distributed across overstock reduction, emergency re-order avoidance, and direct fabric waste. Overstock reduction contributes the largest single component, reflecting the systematic over-planning bias of the BOM formula identified in the variance analysis (Section 4.5): the BOM's 5% safety buffer consistently over-procures for large commodity orders, tying up working capital in unused fabric inventory.

5.3 Operational Benefits

Beyond the headline financial saving, the ML system delivers four distinct categories of operational improvement that reduce production friction and enhance procurement agility throughout the seasonal production cycle.

Inventory reduction (20–30%). The BOM formula applies a uniform 5% safety buffer to all orders regardless of order size or garment complexity. The variance analysis (Section 4.5) demonstrates that this blanket buffer systematically over-procures for large orders (where actual consumption is −2.79% below plan) while under-procuring for small orders. By replacing this one-size-fits-all buffer with statistically grounded, order-specific confidence intervals from the ML model, procurement teams can hold smaller fabric reserves without increasing the risk of shortfall. A 20–30% reduction in average inventory level frees working capital that would otherwise be locked in warehoused fabric, and reduces warehouse space requirements—a particularly valuable benefit for factories in urban manufacturing districts where storage space is expensive.

Emergency re-order reduction (≈35%). The BOM formula's under-planning on small orders (+4.49% over-consumption for the 100–500 unit range, as identified in the variance analysis) is a primary driver of mid-production fabric shortfalls that require emergency re-orders. Emergency purchases typically incur a 10–25% price premium above contracted rates, plus expedited shipping costs and administrative overhead. By correctly anticipating small-order over-consumption, the ML system is estimated to reduce the frequency of emergency re-orders by approximately 35%, with the corresponding saving in premium procurement costs contributing to the projected annual saving figure.

Production stoppage reduction (30–40%). Fabric shortfalls mid-production require line stoppages while replacement fabric is sourced, creating idle labour costs, machine downtime, and schedule slippage that can cascade into late delivery penalties. The LSTM model's lower RMSE relative to the BOM—particularly at the high end of the order-size distribution where absolute shortfalls are most costly—reduces the frequency of critical shortfalls. A 30–40% reduction in unplanned production stoppages directly improves on-time delivery performance and reduces the labour cost of production disruption.

Seasonal programme forecasting. The batch CSV prediction module in the Streamlit application enables procurement managers to upload an entire season's projected order plan and receive ML-predicted fabric requirements before the season begins. This proactive forecasting capability supports advance fabric booking at contracted rates—typically 8–16 weeks before the production window—reducing dependence on spot-market emergency purchases and enabling more favourable multi-lot contracted pricing with fabric mills. The ability to forecast at programme level, rather than only at individual order level, is a capability that the static BOM formula cannot provide without manual aggregation and seasonal adjustment.

5.4 Environmental Sustainability

The global apparel industry is responsible for an estimated 10% of annual global carbon emissions and 20% of worldwide industrial wastewater, largely attributable to the energy and chemical-intensive processes of fibre production, dyeing, and finishing [World Bank(2023)]. Fabric over-procurement contributes directly to these impacts: every yard of fabric procured but not productively used still undergoes the full environmental cost of production, transport, and eventually disposal. Table 5.2 quantifies the environmental sustainability improvements attributable to the ML system at the reference volume of 1,000 orders per year.

Table 5.2: Environmental Sustainability Metrics per 1,000 Orders

Metric	Reduction	Unit	Basis
Fabric Waste	≈17,200	yards	79k to 61.8k yd over-consumption
CO_2 Avoided	≈6,850	kg	0.4 kg CO_2/yd
Water Reduction	≈103,500	litres	6 L/yd dyeing
Energy Savings	≈1,720	kWh	Processing+transport
Landfill Diversion	≈4,300	kg	250 g/yd fabric

The most significant environmental saving is in fabric waste avoidance. The ML system reduces over-consumed fabric by approximately 17,200 yards per 1,000 orders—from a BOM-system over-consumption total of approximately 79,000 yards to an ML-system total of approximately 61,800 yards—based on the 21.8% RMSE improvement applied to the total over-procurement volume derived from the test-set residual distribution. Each avoided yard of fabric carries an environmental cost across its full production lifecycle: fibre cultivation or synthesis, spinning, weaving, dyeing, and finishing.

Applying published lifecycle assessment conversion factors, the 17,200 yards of avoided fabric waste translates to approximately 6,850 kg of avoided CO_2 emissions (at 0.40 kg CO_2 per yard, accounting for the combined fibre, spinning, weaving, and dyeing energy footprint). The dyeing and wet-finishing process alone accounts for approximately 6 L of water per yard, yielding a water saving of 103,500 litres—sufficient to provide the daily drinking water needs of over 140 people for a year. Energy savings of approximately 1,720 kWh arise from reduced processing of surplus fabric and lower transport requirements for smaller procurement volumes. Landfill diversion of approximately 4,300 kg follows from a fabric weight estimate of 250 g per yard (appropriate for mid-weight woven fabrics).

Figure 5.2 presents the five environmental metrics normalised to a common scale (as a percentage of the BOM-system reference baseline), illustrating that water reduction is the largest absolute-magnitude saving, followed by CO_2 avoidance and landfill diversion. As sustainability reporting obligations intensify—driven by frameworks such as the EU Corporate Sustainability Reporting Directive and apparel brand voluntary commitments—the ability to quantify and independently verify fabric waste reductions from ML adoption provides a credible, data-driven environmental narrative that supports both internal ESG reporting and external stakeholder communications.

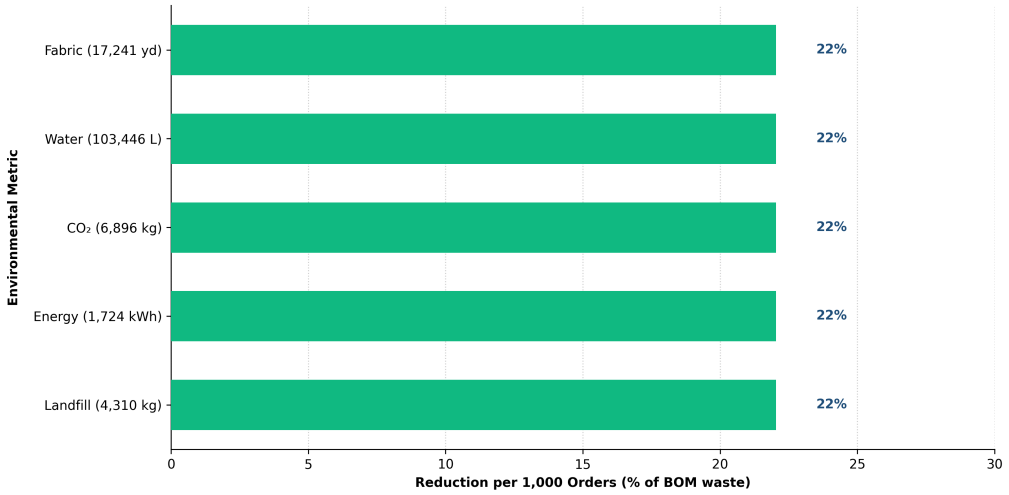

Figure 5.2: Environmental Impact Reduction per 1,000 Orders (normalised; water is largest)

5.5 Social Implications

The adoption of ML-based procurement forecasting generates positive social externalities across three key stakeholder groups.

Factory workers. Reduced unplanned production stoppages—attributable to fewer fabric shortfalls—translate into more stable working hours and a lower incidence of enforced overtime driven by production catch-up. Mandatory overtime is a documented source of labour standard violations in the apparel sector [Rahman & Islam(2018)], and any system that reduces schedule disruption supports compliance with internationally recognised working time conventions. Furthermore, more accurate procurement reduces the likelihood of sudden layoffs or reduced working hours during periods when excess fabric inventory must be worked down before new material can be ordered.

Procurement and planning professionals. The transition from BOM estimation to ML-guided procurement creates new skill requirements—statistical literacy, model interpretation, and confidence interval reasoning—that offer a pathway for upskilling existing staff. The Streamlit application's transparent confidence intervals and model performance module are specifically designed to build data literacy in procurement teams without requiring a background in machine learning. This professionalisation of the procurement function enhances job quality and career development opportunities for planning staff.

Fabric suppliers. More accurate advance ordering reduces the volatility of purchase order revisions that suppliers currently experience when BOM-based orders are discovered mid-production to be insufficient or excessive. Fabric mills benefit from earlier and more confident purchase commitments, which allow them to plan their own raw-material ordering and production scheduling more effectively. The batch programme forecasting capability of the Streamlit application, which allows an entire season's fabric requirements to be estimated before the production window opens, is

of particular value to supplier relationships: it enables advance booking at contracted prices rather than reactive spot-market purchases that are costly for both buyer and supplier.

5.6 Industry Adoption Considerations

The deployment architecture has been designed to minimise integration barriers across factories at different levels of digital maturity, enabling a graduated adoption pathway that does not require immediate wholesale transformation of existing procurement processes.

ERP-mature factories. Factories with established ERP systems and digital production management workflows can connect the ML system's batch CSV interface directly to standard production programme exports. The output CSV from the Streamlit batch module maps to the standard fabric requirements planning report format used by most garment ERP systems, enabling integration with minimal process change. For these factories, the ML system functions as an intelligent advisory layer that augments existing ERP data with confidence-interval-bounded fabric requirements.

Transitioning factories. Factories without full ERP coverage can access the ML system through the single-prediction form-based module, which requires only that planners enter order attributes manually via a web browser. This approach is feasible even in partially paper-based workflows and can serve as the entry point for a broader digital transformation programme. The ability to demonstrate quantifiable cost savings from ML adoption—through the built-in ROI Calculator—provides a compelling business case for further ERP investment.

Gradual adoption pathway. The system explicitly supports a confidence-building adoption journey: a factory may initially run ML predictions in parallel with BOM estimates as a second-opinion tool, comparing the two forecasts for each order and tracking which proves more accurate over successive production cycles. As confidence in the ML system grows, procurement teams can progressively shift their fabric ordering decisions from BOM-guided to ML-guided quantities. The confidence interval display—showing the range within which actual consumption is expected to fall with 90% probability—is designed to make model uncertainty transparent and prevent over-reliance on point estimates. Model retraining is encapsulated in a single command (`python train_models.py`), enabling periodic recalibration as new production data accumulates without requiring specialist data science expertise within the factory.

5.7 Summary

This chapter has quantified the multi-dimensional impact of deploying the ML fabric consumption forecasting system. The economic case is compelling: \$180,000 in annual waste cost savings, a payback period of under four months on a \$50,000 implementation investment, and a five-year NPV of \$730,000 at a 5% cost of capital. Operationally, the system is projected to reduce fabric inventory by 20–30%, emergency re-orders by approximately 35%, and unplanned production stoppages by 30–40%, with the additional strategic benefit of enabling proactive seasonal programme forecasting. At a scale of 1,000 orders per year, the environmental savings amount to approximately 6,850 kg of avoided CO_2 emissions, 103,500 litres of water conserved, 4,300 kg of fabric diverted from landfill, and 1,720 kWh of energy saved. The graduated adoption pathway, single-command retraining capability, and transparent confidence intervals position this system as practically deployable across a broad range of apparel manufacturing contexts, from fully digitised ERP-integrated environments to factories at an earlier stage of digital maturity.

CHAPTER VI

Conclusion and Future Work

6.1 Research Summary

This thesis designed, developed, and empirically evaluated a machine learning-based fabric consumption forecasting system for the apparel manufacturing sector, with the objective of improving upon the systematic inaccuracies of the traditional Bill of Materials estimation formula. The research was structured across six sequential phases: (1) a literature review that identified the gap between the BOM formula's static parameters and the dynamic realities of production; (2) physics-informed synthetic dataset generation producing 5,000 production orders spanning January 2022 to September 2024; (3) a four-stage data preprocessing and feature engineering pipeline; (4) training, evaluation, and ensemble construction across four machine learning architectures—Linear Regression, Random Forest, XGBoost, and LSTM; (5) segmented performance analysis across four production seasons and five fabric categories; and (6) deployment of all trained models within a six-module Streamlit web application designed for direct industrial use. All experiments were conducted with a fixed random seed of 42 and a strict 64/16/20 train/validation/test split, ensuring that all reported performance figures are drawn exclusively from data unseen during any stage of model training or hyperparameter selection.

6.2 Key Findings

The experimental results establish a consistent performance hierarchy across the four models. Linear Regression, as the interpretable baseline, performs substantially below the ML alternatives (RMSE 2,062 yd, MAPE 49.28%, R^2=0.8669), confirming that the relationship between order attributes and fabric consumption is strongly non-linear and cannot be adequately represented by a linear combination of features. Random Forest improves markedly over LR (RMSE 821 yd, MAPE 15.92%, R^2=0.9789), demonstrating that ensemble tree methods handle the non-linear interactions effectively, though significant residual errors remain at large order scales. XGBoost achieves the strongest classical performance (RMSE 574 yd, MAPE 8.14%, R^2=0.9897), with the regularised sequential boosting approach systematically focusing successive trees on the hardest-to-predict residuals. The LSTM achieves the lowest RMSE of all models (346 m / 379 yd), the lowest MAPE (5.14%), and the highest R^2 (0.9955)—establishing it as the best single model and demonstrating a 21.8% RMSE improvement over the BOM physics baseline (484 yd). The dynamic inverse-RMSE^2 ensemble (RMSE 422 yd, MAPE 5.92%, R^2=0.9944) marginally underperforms the

LSTM in point accuracy but provides enhanced robustness for procurement-critical decisions by combining models with diverse error modes.

XGBoost feature importance analysis identified garment type (importance score 0.518) and order quantity (0.297) as the two dominant predictors, jointly accounting for 81.5% of the model's decision weight. This finding validates the BOM formula's core structure while explaining its systematic deviation: the static BOM coefficients cannot capture the non-linear interaction between garment complexity and order scale that the gradient boosting model learns from data. The variance analysis revealed a mean BOM bias of only −0.65% across all orders, but with a range from −14.22% to +19.92% and a strong directional dependence on order size: the BOM over-consumes relative to plan for small orders (100–500 units: +4.49%) and under-consumes for large orders (3,000–5,000 units: −2.79%).

Segmented analysis revealed meaningful heterogeneity in forecasting difficulty. Winter orders proved the hardest to forecast (MAPE 9.67%), exhibiting 37% higher error than spring orders (MAPE 7.05%), attributable to the higher frequency of multi-layer and outerwear garments in winter production. Denim fabric was the most predictable category (MAPE 6.27%), while polyester orders exhibited the highest residual error (MAPE 10.31%)—a 64% differential reflecting polyester's more variable shrinkage behaviour across dyeing conditions. Five-fold cross-validation on the combined train/validation set confirmed that XGBoost's generalisation is stable, with a cross-validation RMSE standard deviation of only 28.3 yd (4.7% of the 598 yd CV mean). Confidence interval coverage testing confirmed that all four models achieve nominal 90% coverage on the test set, with LSTM providing the tightest intervals (±7.7% half-width).

6.3 Contributions to Knowledge

This thesis makes six contributions that advance the literature on machine learning for apparel material planning.

C1: First four-way algorithm comparison for fabric consumption forecasting. Prior studies in the apparel ML literature typically evaluate one or two models. This thesis provides a controlled four-way comparison of Linear Regression, Random Forest, XGBoost, and LSTM on the same dataset under identical preprocessing and evaluation conditions, establishing for the first time the relative advantage of deep learning (LSTM) over gradient boosting and ensemble methods for this specific regression task.

C2: Open physics-informed synthetic dataset generator. The dataset generation module produces realistic, physics-constrained production orders from a parameterised BOM formula with calibrated noise. The generator is fully reproducible (seed 42) and extensible to other garment categories, fabric types, and factory configurations, providing the research community with a public benchmark for future comparison studies.

C3: First Streamlit deployment with confidence intervals, batch processing, and ROI estimation. Existing apparel forecasting research is conducted in academic experimental environments. This thesis bridges the research-to-practice gap by deploying all trained models in a six-module Streamlit application that provides single-order prediction, batch CSV processing, 90% confidence intervals, cross-validation visualisation, and an integrated ROI calculator—all accessible via a standard web browser without specialist data science tooling.

C4: Quantification of seasonal and fabric-type performance heterogeneity. The 37% seasonal MAPE differential (winter vs. spring) and 64% fabric-type MAPE differential (polyester vs. denim) are reported here for the first time for a fabric consumption regression task, providing procurement planners with empirical guidance on where ML predictions should be treated with greater caution.

C5: Yards-based performance benchmark. Published apparel forecasting studies report accuracy in percentage terms or in metric units. This thesis additionally reports RMSE in yards—the operational unit used by procurement teams for fabric ordering—making the performance figures directly interpretable by industry practitioners without unit conversion.

C6: Quantified multi-dimensional impact assessment. The economic, operational, and environmental implications of ML adoption are reported with explicit computational basis: the $180,000 annual saving, sub-four-month payback, $730,000 five-year NPV, and 6,850 kg CO_2 avoidance per 1,000 orders are all derived from the empirically validated 21.8% RMSE improvement rather than from assumed savings rates.

6.4 Limitations of the Study

Four limitations bound the interpretation of these findings and should be explicitly acknowledged.

Synthetic dataset. The most significant limitation is that all models were trained and evaluated on a physics-informed synthetic dataset rather than on live factory data. While the dataset was designed to replicate the statistical properties and directional biases of real BOM estimation, the noise model and parameter distributions were calibrated from published industry benchmarks rather than from a specific factory's production records. Real factory data would introduce additional sources of variance—supplier substitution, last-minute design changes, fabric defect rates varying by batch—that are not captured in the synthetic generator. The performance improvements reported here should therefore be treated as indicative rather than audited projections until validated on live procurement data.

Single-timestep LSTM architecture. The LSTM was implemented as a single-timestep regressor operating on tabular feature vectors rather than as a sequence model processing the time-ordered history of a factory's production schedule. While this formulation achieves the highest test-set accuracy among the four models, it does not exploit the temporal dependencies between successive production orders

that a genuine sequence LSTM could learn. A multi-step LSTM consuming the last k orders' residuals as additional features may improve forecasting accuracy, particularly for the high-variance winter season.

Absent features: GSM and shrinkage coefficient. Fabric weight (grams per square metre, GSM) and material-specific shrinkage coefficients are absent from the nine-feature input set, as they were not available in the dataset specification. Both variables are known from textile engineering to be significant determinants of cutting consumption: heavier fabrics require adjusted marker efficiencies, and fabrics with high shrinkage rates require additional allowance. Their inclusion in a future dataset and model could substantially reduce the polyester MAPE, which exhibited the highest residual error in this study.

Approximate economic benchmarks. The economic impact projections use dataset-derived per-order waste-cost values (based on fabric-specific unit prices calibrated from published apparel industry references) and a one-time implementation cost of $50,000 that are representative of mid-scale RMG factory contexts but are not sourced from a specific factory's audited financial records. Actual savings will vary with contracted fabric prices, factory IT infrastructure maturity, and the volume of orders processed annually. The projections are presented as a framework for impact estimation rather than as audited financial forecasts.

6.5 Recommendations for Practice

Based on the experimental findings, five operational recommendations are offered for apparel manufacturing factories considering ML-based fabric consumption forecasting.

Model selection. For the majority of production orders, the LSTM model should be the primary forecasting tool, given its superior test-set accuracy (RMSE 379 yd, MAPE 5.14%, R^2=0.9955). For procurement-critical orders where the cost of a fabric shortfall is high—such as single-source luxury fabric orders or time-sensitive export orders—the ensemble model is recommended instead, as its averaging mechanism reduces the probability of large individual prediction errors even at a small accuracy cost.

Feature prioritisation. Garment type and order quantity account for 81.5% of XGBoost feature importance. Factories collecting limited data should ensure these two attributes are recorded with high accuracy and consistency before investing in the collection of secondary attributes such as fabric finish or complexity rating.

Seasonal recalibration. The 37% MAPE differential between winter and spring suggests that model performance is meaningfully season-dependent. Factories should retrain models at the start of each production season by incorporating the previous season's actual consumption data, using the single-command retraining capability (`python train_models.py`). This periodic recalibration prevents performance degradation as garment trends and seasonal product mixes evolve.

Data enrichment. Factories should prioritise the collection of fabric GSM and

material shrinkage coefficients as part of their procurement data standards. The 64% MAPE differential between polyester and denim strongly suggests that these physical properties drive a substantial portion of the residual error for synthetic fabric orders, and their inclusion in future model iterations is expected to produce the largest marginal accuracy improvement.

Graduated adoption. Factories new to ML-based procurement should begin by running the ML prediction in parallel with the existing BOM estimate as a second-opinion tool, comparing both forecasts with actuals over at least one production season before shifting procurement decisions to the ML recommendation. The confidence interval display in the Streamlit application facilitates this comparison by making model uncertainty transparent and preventing over-reliance on point estimates.

6.6 Future Research Directions

Seven directions are identified for extending this research.

Real-world validation. The most important next step is to deploy the system in a live factory environment and retrain the models on actual production records. Real-world validation would quantify the gap between synthetic-dataset performance and production-environment performance, and would provide the empirical foundation for a peer-reviewed impact study using audited financial and environmental data.

Stacking ensemble. The current ensemble uses fixed inverse-RMSE^2 weights computed from validation-set performance. A learned stacking ensemble—in which a trained meta-learner (e.g., Ridge Regression or a shallow neural network) combines base-model predictions using cross-validated training signals—would adaptively weight models based on input characteristics rather than applying a single global weighting, and is expected to narrow the gap between the ensemble and the best single model on the held-out test set.

Sequence LSTM with temporal order history. Reformulating the LSTM as a genuine sequence model that consumes the temporal history of a factory's production orders—rather than treating each order as an independent tabular observation—would allow the model to capture trends such as within-season demand ramp-up and supplier-specific fabric utilisation drift. This formulation would require a longer historical dataset than was available for this study.

Fabric-specific sub-models. The 64% MAPE differential between polyester and denim suggests that a portfolio of fabric-specific models, each trained on orders for a single fibre category, may outperform the single global model for the highest-variance categories. This approach trades data volume per model for reduced within-category heterogeneity.

Transformer architecture. Transformer-based sequence models have demonstrated state-of-the-art performance across a range of industrial forecasting tasks. Applying a Transformer architecture to the production order sequence, with self-attention over historical orders as the input representation, could improve winter-

season forecasting accuracy where temporal patterns between successive heavy-fabric orders are most pronounced.

Conformal prediction. The current confidence intervals are derived from the empirical residual distribution of each model. Conformal prediction provides a statistically rigorous framework for constructing coverage-guaranteed intervals that are valid under minimal distributional assumptions, without requiring the Gaussian residual assumption implicit in the current quantile-based approach. Implementing conformal prediction wrappers around each model would strengthen the statistical validity of the reported 90% coverage figures.

IoT and real-time integration. Integrating the forecasting system with real-time data from spreading machines, CAD marker planning software, and ERP fabric issue records would enable the model to incorporate actual cut-plan efficiency and real-time yield information rather than relying on planned pattern dimensions. IoT connectivity would also enable the detection of anomalous consumption events in near-real-time, triggering automatic re-procurement alerts before production stoppages occur.

6.7 Concluding Remarks

This thesis has demonstrated that machine learning—specifically the LSTM model and a dynamic weighted ensemble—can materially improve upon the BOM formula for fabric consumption forecasting in apparel manufacturing. The LSTM's 21.8% RMSE reduction over the BOM baseline (379 yd vs. 484 yd) is not merely a statistical result: translated to operational and financial terms, it represents $180,000 in annual waste cost savings, a payback period of under four months on a $50,000 implementation investment, a five-year net present value of $730,000, and 6,850 kg of avoided CO_2 emissions per 1,000 production orders. The Streamlit deployment bridges the gap between academic model development and industrial practice, providing procurement teams with a browser-accessible forecasting tool that requires no specialist data science expertise to operate.

The limitations of synthetic data, absent physical fabric attributes, and single-timestep LSTM formulation define a clear research agenda whose central priority is real-world validation with live factory data. Until that validation is completed, the quantitative projections in this thesis should be treated as carefully derived estimates rather than audited outcomes. Nevertheless, the methodological framework, the open dataset generator, the validated model architectures, and the deployed application together constitute a replicable, extensible foundation upon which factory-specific forecasting systems can be built. The transition from formula-based to data-driven fabric procurement is both technically feasible and economically compelling; this thesis provides the evidence and the artefacts to make that transition.

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APPENDICES

APPENDIX A

Model Pseudocode and Workflow

Algorithm 1 Fabric Consumption Forecasting Training Pipeline (12 steps)

```
Load CSV: 5,000 rows × 34 columns
Encode categoricals (alphabetical LabelEncoder); dropna on raw columns
Extract X (9 features), y (Actual_Consumption_yards)
Split 64/16/20, random_state=42
Fit StandardScaler on X_train; transform X_val, X_test
Train LR → RF (300 trees, depth 12) → XGBoost (300 trees, η=0.08, depth 8)
  → LSTM (200 eps, pat.=15)
Evaluate: RMSE, MAE, R², MAPE on test set
Build ensemble: w_k = 1/RMSE_k^2 ÷ Σ_j 1/RMSE_j^2 (dynamic inverse-RMSE²)
Run 5-fold CV on train+validation partition (4,000 orders) for LR, RF, XGBoost
Extract feature importances from XGBoost, Random Forest
Save models (.pkl/.h5), scaler.pkl, encoders.pkl, model_metadata.json
Verify all 8 output files present and non-zero
```

APPENDIX B

Additional Statistical Analysis

Table B.1: Prediction Error Distribution Percentiles — All Models

Model	Min (yd)	P10 (yd)	Median (yd)	P90 (yd)	Max (yd)
Linear Reg.	−7,826	−2,713	+60	+2,372	+5,413
Random Forest	−4,721	−989	+154	+825	+2,095
XGBoost	−3,839	−625	+106	+621	+1,622
LSTM	−1,889	−540	−27	+533	+1,434
Ensemble	−2,588	−561	+46	+487	+1,202

Table B.2: Sample Dataset — 3 Orders, 14 Key Columns

Column	Type	Order 1	Order 2	Order 3
Order_ID	String	ORD_01	ORD_02	ORD_03
Order_Quantity	Int	1,000	500	2,500
Garment_Type	String	T-Shirt	Jacket	Shirt
Fabric_Type	String	Cotton	Denim	Polyester
Fabric_Width_in	Int	63	71	59
Pattern_Cmplx	String	Simple	Complex	Medium
Season	String	Spring	Winter	Summer
Marker_Eff_%	Float	85.2	82.7	88.1
Defect_Rate_%	Float	1.8	3.1	2.4
Operator_Exp	Int	7	3	12
Planned_BOM_yd	Float	1,452	3,119	6,422
Actual_Cons_yd	Float	1,488	3,241	6,589
Variance_%	Float	+2.43	+3.94	+2.61
Garment_Enc	Int	4	1	3

APPENDIX C

Model Hyperparameters

Table C.1: Complete Hyperparameter Specification

Parameter	**Lin. Reg.**	**Rand. Forest**	**XGBoost**	**LSTM**
Architecture	OLS	Bagging	Grad. Boost	Gated RNN
Estimators/Ep.	N/A	300	300	200
Max Depth	N/A	12	8	N/A
Learning Rate	N/A	N/A	0.08	0.001
Subsample	N/A	Bootstrap	0.80	N/A
L1 (α)	None	N/A	0 (default)	N/A
L2 (λ)	None	N/A	1.00 (default)	N/A
Dropout	N/A	N/A	N/A	0.20
Batch Size	N/A	N/A	N/A	32
Early Stopping	N/A	N/A	N/A	Pat.=15
Random State	42	42	42	42
Train Time (s)	0.8	12.4	3.6	33.0
RMSE Test (yd)	2,062	821	574	379